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Qualitative financial modelling in fractal dimensions

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Abstract

The Black–Scholes equation is one of the most important partial differential equations governing the value of financial derivatives in financial markets. The Black–Scholes model for pricing stock options has been applied to various payoff structures, and options trading is based on Black and Scholes' principle of dynamic hedging to estimate and assess option prices over time. However, the Black–Scholes model requires severe constraints, assumptions, and conditions to be applied to real-life financial and economic problems. Several methods and approaches have been developed to approach these conditions, such as fractional Black–Scholes models based on fractional derivatives. These fractional models are expected since the Black–Scholes equation is derived using Ito's lemma from stochastic calculus, where fractional derivatives play a leading role. Hence, a fractional stochastic model that includes the basic Black–Scholes model as a special case is expected. However, these fractional financial models require computational tools and advanced analytical methods to solve the associated fractional Black–Scholes equations. Nevertheless, it is believed that the fractal nature of economic processes permits to model economical and financial markets problems more accurately compared to the conventional model. The relationship between fractional calculus and fractals is well-known in the literature. This study introduces a generalized Black–Scholes equation in fractal dimensions and discusses its role in financial marketing. In our analysis, we consider power-laws properties for volatility, interest rated, and dividend payout, which emerge in several empirical regularities in quantitative finance and economics. We apply our model to study the problem of pricing barrier option and we estimate the values of fractal dimensions in both time and in space. Our model can be used to obtain the prices of many pay-off models. We observe that fractal dimensions considerably affect the solutions of the Black–Scholes equation and that, for fractal dimensions much smaller than unity, the call option increases significantly. We prove that fractal dimensions are a powerful tool to obtain new results. Further details are analyzed and discussed.

Keywords: Fractal dimensions, Black–Scholes equation, Power-laws, Option prices

Mathematics Subject Classification: 28A80, 35B08, 35Q40, 35Q91

Introduction

Research on fractal analysis has expanded into various physical and engineering domains since it was introduced by Mandelbrot (Mandelbrot 1982). A fractal is an object characterized by self-similarity and irregularity. This means that the object does not have a typical scale but is characterized by symmetric geometrical properties. Its structural fragments recur through certain intervals in space, but their manifestations and shapes remain intact regardless of the scale at which they are studied (Erokhin and Roshka 2018). Fractals are also described by irregular and non-continuously differentiable functions as their associated graphs are not continuously differentiable at any point. Additionally, they are characterized by a fractional or fractal dimension, which is a pointer of curve complexity. The fractal dimension offers an exceptional way of understanding and characterizing patterns that are frequently found in nature, and it is seriously considered in merely all fields of sciences and engineering. The fractal dimension serves as a practical tool for quantifying the complexity and self-similarity of fractal objects. It has been used to analyze financial data, predict market trends, identify patterns, and test the impact of interest rates, *exchange rate*s, and *inflation* on stock-market performance. Financial fraud detection, which involves the use *of advanced technologies and analytical tools to identify irregular patterns and* potential threats, can also be studied by means of fractal analysis. In the coinage market, the financial system's dimension can be characterized by the state of flux of price quotations. Several studies have considered the importance of fractal dimensions in finance mainly in studying financial and statistical time series (Gneiting et al 2012; Sánchez-Granero et al. 2008, 2012a, b) where fractal dimension is correlated with the Hurst exponent (a fractal parameter of an economic process), which is frequently used in financial analysis to study long-memory effects in a series of stock prices. Stock prices at a small and large time scale share a common self-similarity property determined by the Hurst exponent. This is connected to the Fractal Market Hypothesis (Mandelbrot and Hudson 2006), which offers a plausible explanation of various market inefficiencies and confirms the fact that "only balance of long-term and short-term players on the market can assure the market effectiveness." Potential divergence from the equilibrium is displayed on the Hurst exponent. Fractional Brownian motion (FBm), which belongs to a class of stochastic volatility models, is also used as an alternative to study long-memory effects in finance, mainly for the logarithm of the price of a stock, which is generally unpredictable (Niu and Liu 2010; Rostek and Schobel 2013). A new fractional parameter is introduced in this model, which is advantageous as it helps to describe the option market's behavior. Unlike many other stochastic volatility models, the fractional model provides a natural extension to the classical Black–Scholes model.

Other methodologies use the concept of Levy stable fractional motions to describe the evolution of a stock's price, taking volatility into account (Kim et al. 2019; Panas 2001). Several studies have demonstrated that the fractal dimension is not the inverse of the self-similarity index (Fernández-Martínez, 2019). Particularly, for a wide range of random and stochastic processes, such as the fractional Brownian motions and fractional Levy stable motions, the fractal dimension is related to the Hurst exponent. This is an interesting result as it provides confidence in using fractal dimensions to study financial stocks, although their associated dynamical models are more complicated than the basic ones. In general, fractional financial models are motivating as they add thoughtfulness to the nature of economic systems. Additionally, the fractal aspects of the financial process allow to accurately model several markets characterized by anomalous market returns. Price option evolution is a highly complicated aspect in financial markets owing to its random aspects; that is, prices are random and exhibit an irregular trend. However, its complexity comprises self-similar patterns or fractals. These offer an improved quantitative control and help taking a better qualitative financial running and asset decisions (Blackledge 2010).

In financial markets containing derivative investment instruments, the Black–Scholes mathematical model plays a central role mainly in European-style options (Blackledge 2022). Black and Scholes assume that the stock price follows a random stochastic walk, with a constant mean and variance of the rate of return. Since the call price is naturally expected to be an increasing function of stock price, one can construct a theory of option pricing derived from the absence of profitable arbitrage (i.e., no arbitrage). Hence, the rate of return on this portfolio must be risk-free return. The Black–Scholes equation is deduced from a parabolic partial differential equation and stochastic arguments and is practical, since, given the risk of the security and its predictable return, it gives an estimate of the price and shows that the option has a unique price. The value of the call at the expiration date gives a boundary condition required to solve the Black–Scholes partial differential equation. The basic idea is to perform stochastic analysis backwards in time from the value at expiration to find the call price at earlier times. Remarkably, the call price rises as the time to expiration increases, and hence, *the call option* becomes less valuable as the strike *price increases*; the BSE does satisfy this property. Moreover, if the stock price is expected to increase, then purchasing the call will cost far less than purchasing the stock. The increase in interest made the grouping of calls and risk-free bonds more attractive (i.e., the call price will tend to increase with increases in interest rates). A higher variance tends to increase the call price, which is obvious for an out-of-the-money option but also holds for an in-the-money option asa higher variance also increases the potential for a big profit. We stress that the mean rate of return (or expected return) on the stock has no effect on the call price.

However, several observations predict that the price dynamics of stock markets is characterized by a long-range dependence in asset prices. In fact, the Black–Scholes model does not take this dependence into account (Blackledge and Lamphiere 2022; Salopek 1998). The fractional stock market prices model, in which arbitrage opportunities exist under a complete and frictionless setting, offers a potential solution mainly when the modulating process is a fractional Brownian motion (Dai and Heyde1996; Ichiba et al. 2021). Both the fractal structure for financial market and fractional formulation of the Black–Scholes model are considered currently active fields of research. The fractional Brownian motion is not a semimartingale, and this property is crucial in modeling stock price and rough volatility, in addition to constructing hedging policies and arbitrage strategies (Black and Scholes 1973).

Recently, fractals have been widely used to observe financial markets and analyze their behavior. They have been used by Verna and Kumar (2024) to study the financial performance of consequential companies after merges and acquisitions and to have a naïve idea about factors that manipulate financial performance and strategies to conquer them. Fractals have been also employed to analyze empirically market efficiency at various assets an time scales (Brouty and Garcin 2024) and examine the impact of corporate governance index on working capital management (Gupta and Pandey 2024). Notably, the efficient market hypothesis (EMH) is not a convincing model of financial markets; it has been replaced by the fractal market hypothesis (FMH) as a more general alternative tool to study financial markets, which are also characterized by a fractal structure in investment horizons (Vacha and Vosvrda 2005). Fractals have been also recently used in analyzing market efficiency and the daily return series of the stock index of NASDAQ (National Association of Securities Dealers Automated Quotations) stock exchange (Arashi and Rounaghi 2022). Investigations have been conducted on the FMH and forecasting time-series stock returns for both Tehran London stock exchanges (Moradi et al 2021). Mosteanu et al. (2019) studied the connections between financial decision, financial performance, and fractals patterns. Various fractal properties of financial markets were obtained by Budinski-Petković et al. (2014), who they proved that fractal functions have great modeling abilities concerning market dynamics. Wu et al. (2021) studied fractal statistical measure and portfolio model optimization under power-law distribution. Moreover, fractal dimensions in finance have been estimated using various approaches, including wavelet analysis (Bayraktar et al. 2004).

Most financial systems rely on the EMH, which includes the Black–Scholes formula for placing an option. However, these systems are flawed since EMH is based on statement that economic processes are normally distributed (Gaussian random walks). This leads to some failure in predicting a market share value's future volatility. Therefore, a new financial risk assessment model is required to remedy this defect. The most confident mathematical tool present nowadays is based on Lévy statistics and a nonstationary fractional diffusion equation characterized by the Lévy index. The fractional aspects of the Black–Scholes model have been discussed widely in literature based on various arguments, such as the time-fractional Black–Scholes equation (hereafter BSE) with the Caputo derivatives (Fall et al. 2019) and the Liouville-Caputo fractional derivative (Sawangtong et al. 2018); fractional functional with two occurrences of integrals (El-Nabulsi 2015); and pricing-fractional BSE with the Weyl fractional derivative (Cartea and del-Castillo-Negrete 2007; Chen et al. 2014) or by combining the Caputo derivative with Weyl derivative simultaneously (Kittipoom 2018). Araneda (2024) introduced a multifractional option pricing formula, and Zhang et al. (2024) reviewed fractional BSE and their solution techniques. Li et al. (2020) conducted a fractional investigation of bank data with fractal-fractional Caputo derivative, and Li et al. (2023) investigated the financial bubble mathematical model under fractal-fractional Caputo derivative. The works of Garcin (2017, 2019, 2020, 2022) reflect the importance of fractional Brownian motion in finance with random exponent. The estimation of volatility using the fractional Brownian motion to model asset returns is an important chapter in qualitative finance. The Hurst exponent is also considered a major parameter, and a method to extract realized Hurst exponents from log-prices was proposed by Garcin (2020). The works of Bianchi et al. (2012, 2013, 2015, Bianchi and Pianese 2015; Bianchi and Frezza 2017) also reflect the relevance of multifractional process with random exponent and fractal stokes markets in financial studies. Additional recent works on the Hurst exponent and volatility are those of Di Persio and Turatta (2022) and Zournatzidou and Floros (2023). Mattera and Di Sciorio (2021) studied option pricing under multifractional process and long-range dependence by introducing a new method to compute the European Call (and Put) option price under the assumption of multifractional Brownian motion. Despite the benefits of these methodologies in addition to the approaches using fractional derivatives operators, analyses of the BSE in fractal dimensions are lacking, even though arbitrage in fractal modulated markets is well-known in the literature (Peters 1994),and a fractal version of BSE was constructed by Cutland et al. (1995). This is the main aim of the present study.

Attempts have been made to use fractal analysis in financial studies and in estimating financial volatility. Several studies have proved that higher-risk assets are characterized by higher fractal dimensions, whereas stable assets have lower fractal dimensions close to unity. Fractal dimensions proved to be a practical tool as stock market prices are fractal objects, and hence, fractal dimensions are used to measure the degree of complexity (Crisan 2015; Peters 1994). Fractal dimensional analysis has been also used in the study of Indian financial markets (Priyadarshini and Chandra Babu 2012) and to examine the statistical properties of volatility among the New York, Tokyo, Taiwan, South Korea, Singapore, and Hong Kong stock markets (Yu and Huang 2004). Since finance in general belongs to a class of random dynamical systems, we consider fractal dimensions using a new generalized approach based on the local generalized fractal derivative operator.

Fractal derivatives are generally introduced as $\nabla^{\alpha} f(x) := df(x)\big/ dx^{\alpha} = (x^{1-\alpha}\big/\alpha)\nabla f(x), \alpha > 0$, assuming that for a given function f, the derivative $df\big/dx$ and the fractal derivative exist simultaneously. The gradient operator $\nabla f(x)$ is regularly defined on a smooth space, and it becomes unsound for discontinuous space in contrast to the fractal derivative $\nabla^{\alpha} f(x)$ (He 2018). These fractal derivatives are considered an extension of Leibniz's derivative for discontinuous fractal media (El-Nabulsi and Anukool 2023f, 2024a, b; Fan and He 2012; He 2011). They are comparable to those derivatives operators introduced in the literature within the framework of "product-like fractal geometry" (Li and Ostoja-Starzewski 2009, 2011, 2020) used in anisotropic and non-continuum fractal-dimensional media (El-Nabulsi 2021a, b, c, d, e, f, g, h; 2022a, b; El-Nabulsi and Anukool 2021a, b, 2022a, b, c, d, e, f, g, h, i, j; 2023a, b, c, d, e, 2024c; Ostoja-Starzewski 2008, 2009; Ostoja-Starzewski et al. 2014). In fact, Lima (2017) discussed the modeling of the financial market using the anisotropic Ising model, obtaining the mean price in terms of a site anisotropy term, which reinforces the sensitivity of the agent's sentiment to external news.

To take into account fractal dimensions, we generalize this definition based on the concept of gradient and Laplacian operators (Stillinger 1977). According to this approach, the integration of a radially symmetric function on D-dimensional fractional space leads to a D-dimensional fractional Laplacian: $\Delta_D = \nabla_r^2 + ((D-1)\big/r)\nabla_r$, where $\nabla_r = \partial\big/\partial r$, and $0 < D \leq 1$. The fractional-dimensional generalization of ∇_r is done in (Zubair et al. 2010, 2011a, b) where the square of Δ_D is approximated by ∇_D, and leads to $|\nabla_D| \approx \nabla + (D-1)\big/2r, |r| >> 1$. In the present work, we generalize the fractal derivative for the case of fractal dimension by introducing the fractal operator $\overline{\nabla}^{\alpha} f(x) := (x\big/x_0)^{1-\alpha}(\nabla + ((1-\alpha)\big/2x))f(x)$, where x_0 is a spatial-parameter introduced to take into account the dimensionality problem. Moreover, we introduce a generalized, conformable gradient time-operator that takes into account the time-fractal dimension.

In fact, conformable derivatives are local operators that modify the basic integer-order derivatives defined as follows: let $t \in [0, \infty)$ and $f(t) : [0, \infty) \to \mathbb{R}$ be a function. Its conformable derivative of order $\beta \in (0, 1)$ is defined by $T_\beta(f)(t) = \lim_{\varepsilon \to 0} \left(f\left(t + \varepsilon t^{1-\beta}\right) - f(t)\right) / \varepsilon$. If $f(t)$ is β-differentiable in some $(0, a)$, $a > 0$ and $lim_{\varepsilon \to 0} T_\beta(f)(t)$ exists in the limit's positive region, then $T_\beta(f)(0) = lim_{\varepsilon \to 0} T_\beta(f)(t)$ (Khalil et al. 2014). This definition is motivating when compared to the fractional derivative, since it reduces the complexity of the dynamical equations of the system under study. Furthermore, when conformable derivatives are taken into account, one can add extra-parameters into the dynamical equations without deformed their physical properties. This definition is correlated to the classical derivative as follows: $T_\beta(f)(t) = t^{1-\beta} \partial f / \partial t$ (Anderson et al. 2019), and its generalization in the time-fractal dimension is given by $\overline{T}_\beta(f)(t) = (t/t_0)^{1-\beta} (\partial / \partial t + (1 - \beta) / t) f(t)$. t_0 is a time-parameter introduced to take into account the dimensionality problem, and in this study, it is set equal to unity for convenience. In this work, we implement these definitions of fractal derivatives into the BSE and study some of its applications in quantitative finance. Before elaborating on our study, we address a summary table of prior attempts and important studies on the topic. Most studies have focused on fractal, multifractal time series and fractional Brownian motion having fractal properties, rather than on the concept of fractal dimensions presented in this study. Hence, we believe that this study contributes to the literature and aids in understanding the usage of fractals in financial modeling for future study (Table 1).

The remainder of the paper is organized as follows. In the "Construction of the fractal qualitative Black–Scholes model" section, we introduce the basic setups of our fractal model. The "Qualitative and computational analyses" section presents discusses the basic properties of the fractal financial model. The Section "Discussion and robustness analysis" discusses the empirical results. Finally, the Section "Conclusions" concludes the paper and discusses future implications.

Construction of the fractal qualitative Black–Scholes model

Before introducing the BSE in fractal dimensions, we recall the basic setups of stochastic theory. Let $W : \mathbb{R} \times \Omega \to \mathbb{R}$ be a stochastic process—for instance, stock price with Ω is some probability space. A stochastic process $W_t, t \geq 0$ is a Brownian motion if $W_0 = 0$, $W_t - W_s, s < t$ is continuous, with normal distribution and variance $t - s$, and the distribution $W_t - W_s$ is independent of $W_r, r \leq s$ (Turner 2010). Let us suppose that $X_0 \in \mathbb{R}$ and $r, \sigma \in \{F_t^X\}_{t \geq 0}$ where F_t^X denotes the "information generated by S on the interval $[0, t]$." Here $r > 0$ and σ are the risk-free interest rate and volatility, respectively. The following properties hold:

Table 1 Summary of some basic fractal models discussed in literature

Models	Features
The fractal model for currency futures price dynamics (Fang el al. 1994)	The model used a time series having fractal structures characterized by long-term dependence and nonperiodic cycles
The generalized model for asset returns based on Fractal Market Hypothesis (Rachev et al. 1999)	The model takes into account the capital market systems in which the Conditionally Exponential Dependence (CED) property can be attached to each investor on the market. Neither fractal dimensions nor fractal calculus have been used
The fractal model of foreign exchange markets based on the modified rescaled range (R/S) (Mulligan 2000)	The model treats the post-Bretton Woods period using R/S, long memory and fractal structure
The fractal model based on multifractal random walks used to study financial time series (Barcy et al. 2001)	The model used solvable "stochastic volatility" processes
The fractal forecasting model for financial time series (Richards 2004)	The author used the *state transition-fitted residual scale ratio* model to predict the conditional probability of extreme events
The fractal modulated BS models with stochastic volatility (Bayraktar and Poor 2005)	The model is based on fractional Brownian motion which is not a semi-martingale and hence additional constraints are required
Multifractal analysis of Chinese stock volatilities based on volatility partition function approach (Jiang and Zhou 2008)	The model used multifractal analysis on the 1-min of two indexes and 1139 stocks in the Chinese stock markets based on the partition function approach and not on the concept of fractal dimensions
Risk and complex fractals for stock market (Jumarie 2010a)	The model is also based on fractional Brownian motion yet a Black–Scholes equation of order *n* is derived
The model based on "Fractal Market Hypothesis" during financial crisis (Kristoufek 2012)	The model focused on liquidity and investment horizons to study the dynamics of the financial markets during Global Financial Crisis at late 2000s
The fractal model based on time-varying Hurst-Hölder exponents (Bianchi and Pianese 2018)	The model used the idea that financial princes evolve in a continuum of equilibria and disequilibria
The fractal model of foreign exchange rates (Garcin 2020)	The model used fractal analysis of the mutlifractality and proposed a data consistent model for the dynamic of the Hurst exponent
The fractal model based on information theory (Brouty and Garcin Brouty and Garcin 2024)	The model used entropy and fractional Brownian motion having fractal properties
The model based on fractal dimensional analysis of stock prices using fractal interpolation (Verna and Kumar 2023)	The model used fractal interpolating functions at different scale and times and described stock prices fluctuation with different perspectives; yet the model did not treat BSE with time-dependent components

- if S_t satisfies the stochastic differential equation $dS_t = rS_t dt + \sigma S_t dW_t$ with a constant initial value $S_0 > 0$, then $S_{t+h} - S_t - hrS_t - \sigma S_t (W_{t+h} - W_t)$ is a random variable with mean and variance, which are $\mathrm{O}(h)$ (Lyasoff 2017),
- if S_t satisfies the stochastic differential equation $dS_t = rS_t dt + \sigma S_t dW_t$ with a constant initial value $S_0 > 0$, then:

$$S(t) = S_0 + \int_0^t rS_u du + \int_0^t \sigma S_u dW_u, \tag{1}$$

where the 2nd integral is the Itô stochastic integral. An Itô process $X_t, t \geq 0$ satisfies the stochastic differential equation $dX_t = r(t)X_t dt + \sigma(t)X_t dW_t$. Its solution is given by $S(t) = S_0 \exp(\sigma W_t + (r - \sigma^2 t/2))$ and is characterized by a lognormal distribution for $t > 0$. It is a Markov process but not a process of independent increments. Here, r is the risk-free interest rate and σ the volatility. Based on Itô lemma, if we let $f(x, t)$ be a twice-differentiable function, then $f(X_t, t)$ is an Itô process that satisfies the following equation:

$$d(f(X_t, t)) = \frac{\partial f}{\partial t}(X_t, t)dt + \nabla_{X_t} f dX_t + \frac{1}{2}\Delta_{X_t} f dX_t^2, \tag{2}$$

with $dt^2 = 0$, $dtdW_t = 0$ and $dW_t^2 = dt$. Here $\nabla_{X_t} := \partial/\partial X_t$ is the gradient operator, and $\Delta_{X_t} := \partial^2/\partial X_t^2$ is the Laplacian operator. This formalism has led Black and Scholes to formulate their financial stochastic theory based on the following concepts: assuming that the trading strategies of a frictionless market are free from arbitrage opportunities, that the purchase of any quantity of a stock and short-selling is allowed, and that the basic stock does not pay any dividends, then for any stochastic process $Z = V(t, S(t))$ where $V : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}$ is a C^2-function, the following BSE of a European option holds:

$$rS\frac{\partial V}{\partial S} + \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0. \tag{3}$$

Here, S is the current stock or sport price. Using this equation, one can find the price of an option. However, investors are often interested in predicting the future price of an option and hence in creating a gainful portfolio. Equation (3) does the work by describing the price of option over time. Despite its tremendous success in modeling the stock option price, several critiques have been raised regarding its domains of applicability, since, in most financial options, the volatility and risk-free interest rate vary with time (El-Nabulsi and Golmankhaneh 2021; Morales-Bañuelos et al. 2022; Rana and Ahmad 2012; Rodrigo and Mamon 2006). The option pay prices are consistently higher than those predicted by Eq. (3). The better way to solve these issues is by considering a BSE with time-varying parameters with dividend payout $D(t)$ and a price of a call option with a strike price of K. It takes the following form:

$$\frac{\partial V}{\partial t} = (r(t) - D(t))S\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2(t)S^2\frac{\partial^2 V}{\partial S^2} - r(t)V, \tag{4}$$

where $V(T, S)_{t=T} = (S(T) - K)^+ = \max(S(T) - K, 0)$ is the dividend payout. The solution of this equation is obtained after performing the change of variables $V(t, S(t)) = g(t)\overline{V}(\overline{t}, \overline{S}(\overline{t}))$, $\overline{S}(t) = S(t)\varphi(t)$, $\overline{t} = \psi(t)$ which gives, after algebra,

$$V(t, S(t)) = \xi e^{-\int_t^T (r(u) - \zeta\sigma(u))du)\overline{V}(\overline{t}, \overline{S}(\overline{t}))}. \tag{5}$$

Here $\xi = K/\overline{K}$, $\zeta = r_c/\sigma_c^2$, r_c, σ_c^2 and $\overline{K}$ are here the risk-free interest rate, volatility, and strike price respectively (Awasthi and Riyasudheen 2020). The generalization of this approach for the case of $r(S, t), \sigma^2(S, t)$ and $D(S, t)$ has been addressed in several research studies using numerical techniques (Lai 2019; Lin et al. 2021; Mohammadi 2015; Rao 2018; Valkov 2014). Several numerical studies have found that the

RHS of Eq. (4) (denoted here by $L_{BSE}(V)$) is non-zero (Windcliff et al. 2004). This shows that, under firm assumptions, the BSE does not reproduce correctly the concrete market conditions. Several ways may be introduced to remedy this problem—for example, by assuming that the RHS is a second derivative of the option price to the underlying asset prices; in other words, it is $L_{BSE}(V) \propto S^2\partial^2 V/\partial S^2$ based on the assumption that the volatility of the underlying asset is not a constant (Al-Zhour et al. 2019). It is noteworthy that numerical and computational analyses are of particular importance asthey support researchers with a series of financial tests to validate their theories (Golbabai et al. 2019; Golbabai and Nikan 2020; Nikan el al. 2022; Yang et al. 2020; Yavuz 2018).

In this paper, we assume that the modification term coming from $L_{BSE}(V) = \xi\eta(S,t)V$ is a set of discrepancy data related to the underlying option price. ξ here is a control parameter. We generalize this approach using the fractal operators $\overline{\nabla}$ and T_β which generalizes the approaches discussed in Lo and Hui (2011) and Yavuz and Özdemir (2018).We use the following gradient and Laplacian operators:

$$\overline{\nabla}_S := \sqrt{a}S^{1-\alpha}\left(\frac{\partial}{\partial S} + \frac{1-\alpha}{2S}\right), \tag{6}$$

$$\bar{\Delta}_S := \bar{\nabla}_S\bar{\nabla}_S = aS^{2(1-\alpha)}\frac{\partial^2}{\partial S^2} + bS^{1-2\alpha}\frac{\partial}{\partial S} + cS^{-2\alpha}, \tag{7}$$

where

$$a = S_0^{2(\alpha-1)},\ b = \frac{5}{2}(1-\alpha)S_0^{2(\alpha-1)},\ c = \frac{(\alpha-1)(3\alpha-1)}{12}S_0^{2(\alpha-1)}, \tag{8}$$

and S_0 is the current price of the stock at initial time t_0. Interchanging the partial derivative $\partial f/\partial t$ with T_β and $\partial/\partial S$ with $\bar{\nabla}$ in the generalized BSE (4) gives, after algebra, the subsequent modified BSE in fractal dimensions (α,β)—in other words,$(t,S) \to (t^\beta, S^\alpha), (\alpha,\beta) \in (0,1)$:

$$\begin{aligned} A\left(\frac{\partial V}{\partial t} + \frac{1-\beta}{t}V\right) &= \frac{1}{2}a\sigma^2(S,t)S^{4-2\alpha}t^{\beta-1}\frac{\partial^2 V}{\partial S^2} + \left(\sqrt{a}(r(S,t)-D(S,t))S^{2-\alpha} + \frac{1}{2}b\sigma^2(S,t)S^{3-2\alpha}\right)t^{\beta-1}\frac{\partial V}{\partial S} \\ &+ \left(\frac{1-\alpha}{2}\sqrt{a}S^{1-\alpha}(r(S,t)-D(S,t)) + \frac{1}{2}c\sigma^2(S,t)S^{2-2\alpha} - r(S,t)\right)t^{\beta-1}V - \xi\eta(S,t)V. \end{aligned} \tag{9}$$

Here $A = t_0^{\beta-1}$. In the next section, we analyze this equation and study several of its basic properties and features in quantitative finance. The subsequent change of variables:

$$\frac{a}{A}\sigma^2(S,t) = \overline{\sigma}^2(t)S^{2\alpha-2}t^{1-\beta}, \tag{10}$$

$$\xi\eta(S,t)=-r(S,t)t^{\beta-1}+\frac{\beta-1}{At},\tag{11}$$

$$\sqrt{a}(r(S,t)-D(S,t))t^{\beta-1}S^{1-\alpha}=\overline{r}(t)-\overline{D}(t)-\frac{b}{2aA}\overline{\sigma}^2(t),\tag{12}$$

$$\overline{r}(t)=\frac{1-\alpha}{3-\alpha}\overline{D}(t)-\frac{\alpha+1}{3-\alpha}\frac{c}{2aA}\overline{\sigma}^2(t),\tag{13}$$

$$r(S,t)-D(S,t)=\overline{G}(t)S^{\alpha-1},\tag{14}$$

$$\sqrt{a}\overline{G}(t)t^{\beta-1}=\overline{r}(t)-\overline{D}(t)-\frac{b}{2aA}\overline{\sigma}^2(t)=\frac{2}{\alpha-3}\overline{D}(t)-\frac{2}{3-\alpha}\frac{b}{aA}\overline{\sigma}^2(t),\tag{15}$$

convert Eq. (9) to

$$\frac{\partial V}{\partial t}=\frac{1}{2}\overline{\sigma}^2(t)S^2\frac{\partial^2 V}{\partial S^2}+\left(\overline{r}(t)-\overline{D}(t)\right)S\frac{\partial V}{\partial S}-\overline{r}(t)V.\tag{16}$$

The solution of Eq. (16) is given by Lo and Hui (2011) and Lo et al. (2003):

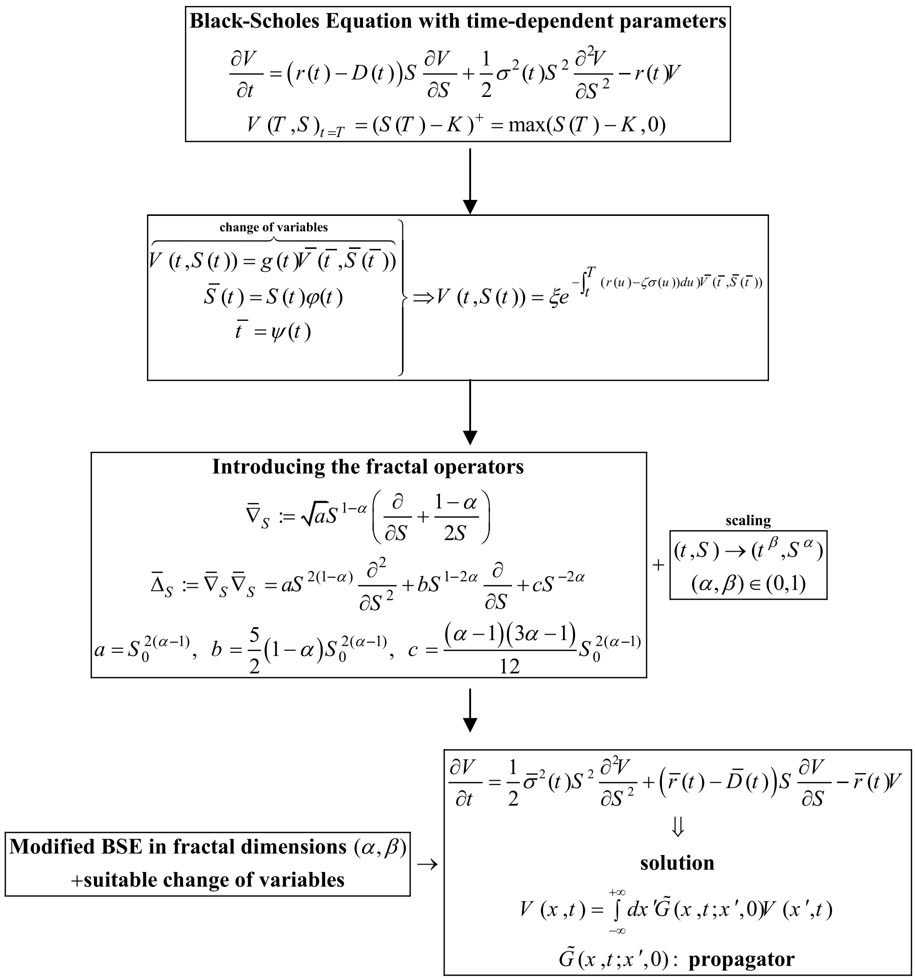
$$V(x,t)=\int_{-\infty}^{+\infty}dx'\tilde{G}(x,t;x',0)V(x',t),\tag{17}$$

where

$$\tilde{G}(x,t;x',0)=\frac{1}{\sqrt{2\pi\int_0^t\overline{\sigma}^2(t')dt'}}\exp\left(-\frac{\left(x-x'+\int_0^t\left(\overline{r}(t')-\overline{D}(t')-\frac{1}{2}\overline{\sigma}^2(t')\right)dt'\right)^2}{2\int_0^t\overline{\sigma}^2(t')dt'}-\int_0^t\overline{r}(t')dt'\right),\tag{18}$$

is the propagator and $x=\ln(S/S_0)$ a new change of variable. In what follows, we suggest the following power-laws: $\overline{\sigma}^2(t)=\overline{\sigma}_0^2t^{\gamma}$, $r(S,t)=r_0t^{\xi}S^{\alpha-1}$ and $D(S,t)=D_0t^{\xi}S^{\alpha-1}$, where γ and ξ are real constants, and $\overline{\sigma}_0^2$, r_0 and D_0 are real parameters. These power-laws play an important role in finance; in fact, several studies offer evidence of the power-law properties of $\overline{\sigma}^2(t)$, $r(S,t)$, and $D(S,t)$.

Let us sketch a schema with the methodology steps used in our approach:



Schema of the methodology used to derive the modified BSE in fractal dimensions and its solution.

Several research works present evidence that time-varying volatility can account for the power law property of high frequency stock returns (Bouchaud 2001; Farmer and Lillo 2004; Gabaix 2009, 2016; Gabaix et al. 2003, 2006; Lillo and Mantegna 2003; Mandelbrot 2001; Takaishi 2020); for instance, time-varying power-law volatility can account for the power-law property of high frequency stock returns (Warusawitharana 2016, 2018). In fact, in option-pricing financial applications, the choice of mathematical form of the variance estimate is more important than the choice of interest rate. The importance of scale-free behavior is well-known in fractals characterized by power-law scaling valid across a given range. To physicists, the obvious occurrence of power-laws in quantitative finance is an important clue to model them. These power-laws may or may not be well-matched with economic equilibrium and stability, but in either case, this suggests a different emphasis. Notably, several statistical properties of financial markets are described asymptotically by power-laws that have motivating consequences for volatility forecasting and derivative pricing. These power-laws offer new insights in financial markets modeling (Lux 2006; Lux and Alfarano 2016). They are supported by high-frequency data, analytical, and numerical tools which hold across dissimilar markets, time periods, and distributions of daily stock volume and returns.

These power-law findings are ubiquitous and important, since information arising from estimate prices with tail-risk sensitivity are particularly critical for applied economists and finance researchers (Nirei et al. 2020). In general, power-laws in volatility and financial returns are correlated, and they are considered one of the main features and basic characteristics of financial data. One of these observed power-laws includes the transaction volume, which is characterized by long-range dependence. Additional power-laws are reported for fluctuations in the Japanese stock market (Takayasu 2002). In general, financial returns are not identically and independently distributed, since no overall trends are observed, and their correlated distributions are free from fluctuations. Hence, they are represented by a stochastic process rather than a time-invariant distribution as financial analysts would like to estimate financial situations involving uncertainties such as volatile markets and inflation rates. In our approach, this gives

$$\overline{r}(t) = (\alpha - 1)\sqrt{a}(r_0 - D_0)t^{\beta-1+\xi} + \left(\frac{c(\alpha+1)}{2} - 2b(\alpha-1)\right)\frac{1}{aA(\alpha-3)}\overline{\sigma}_0^2 t^{\gamma}, \tag{19}$$

$$\overline{D}(t) = (\alpha - 3)\sqrt{a}(r_0 - D_0)t^{\beta-1+\xi} - \frac{2b}{aA}\overline{\sigma}_0^2 t^{\gamma}, \tag{20}$$

$$\overline{G}(t) = (r_0 - D_0)t^{\xi}, \tag{21}$$

We assume, for simplicity, that $\xi = -\gamma = 1 - \beta$—in other words, the variance has alinear term structure. Accordingly, we find that

$$\overline{r}(t) = \underbrace{(\alpha - 1)\sqrt{a}(r_0 - D_0)}_{P_\alpha} + \underbrace{\left(\frac{c(\alpha+1)}{2} - 2b(\alpha-1)\right)\frac{1}{aA(\alpha-3)}\overline{\sigma}_0^2}_{Q_\alpha} t^{\beta-1} = P_\alpha + Q_\alpha t^{\beta-1}, \tag{22}$$

$$\overline{D}(t) = \underbrace{(\alpha - 3)\sqrt{a}(r_0 - D_0)}_{\overline{P}_\alpha} - \underbrace{\frac{2b}{aA}\overline{\sigma}_0^2}_{\overline{Q}_\alpha} t^{\beta-1} = \overline{P}_\alpha - \overline{Q}_\alpha t^{\beta-1}, \tag{23}$$

$$\overline{r}(t) - \overline{D}(t) - \frac{1}{2}\overline{\sigma}^2(t) = \underbrace{\left(P_\alpha - \overline{P}_\alpha\right)}_{A_\alpha} + \underbrace{\left(Q_\alpha + \overline{Q}_\alpha - \frac{1}{2}\overline{\sigma}_0^2\right)}_{B_\alpha} t^{\beta-1} = A_\alpha + B_\alpha t^{\beta-1}. \tag{24}$$

Remark 1 For $\alpha = 1$, we can reduce Eq. (9) to the following modified BSE:

$$A\frac{\partial V}{\partial t} = \frac{1}{2}\overline{\sigma}_0^2 S^2 \frac{\partial^2 V}{\partial S^2} + r_{eff} S\frac{\partial V}{\partial S} - r_{eff} V,$$

if we assume that $\sigma^2(t) = \overline{\sigma}_0^2 t^{1-\beta}$, $r(t) = r_0 t^{1-\beta}$, $D(t) = D_0 t^{1-\beta}$ and $\xi\eta(t) = A(\beta-1)/t$ with $D_0 = 0$. Here $r_{eff} := r_0 + \overline{\sigma}_0^2/2$ is the effective interest rate. These power-laws have been discussed in finance, and several studies have reported that time-varying volatility can account for the power-law property of high-frequency stock returns (Malevergne et al. 2005).

Remark 2 For $\beta = 1$ and for the case of time-independent variables, we can reduce Eq. (8) to the following modified BSE:

$$\frac{\partial V}{\partial t} = \frac{1}{2}a\sigma^2(S)S^{4-2\alpha}\frac{\partial^2 V}{\partial S^2} + \left(\sqrt{a}(r(S) - D(S))S^{2-\alpha} + \frac{1}{2}b\sigma^2(S)S^{3-2\alpha}\right)\frac{\partial V}{\partial S} + \left(\frac{1-\alpha}{2}\sqrt{a}S^{1-\alpha}(r(S) - D(S)) + \frac{1}{2}c\sigma^2(S)S^{2-2\alpha}\right)V - (r(S) + \xi\eta(S))V.$$

The following power-laws $\sigma^2(S) = \overline{\sigma}_0^2 S^{2\alpha-2}$, $r(S) = r_0 S^{\alpha-1}$, $D(S) = D_0 S^{\alpha-1}$ and $\eta(S) = \eta_0 S^{\alpha-1}$ reduce the previous equation to

$$\begin{aligned}\frac{\partial V}{\partial t} =& \frac{1}{2}a\overline{\sigma}_0^2 S^2\frac{\partial^2 V}{\partial S^2} + \left(\sqrt{a}(r_0 - D_0) + \frac{1}{2}b\overline{\sigma}_0^2\right)S\frac{\partial V}{\partial S} \\ &+ \left(\frac{1-\alpha}{2}\sqrt{a}(r_0 - D_0) + \frac{1}{2}c\overline{\sigma}_0^2 - (r_0 + \xi\eta_0)S^{\alpha-1}\right)V.\end{aligned}$$

The change of variable $S = S_0 e^x$ ($S_0 = 1$ for convenience) converts this equation to

$$\begin{aligned}\frac{\partial V}{\partial t} =& \frac{1}{2}\overline{\sigma}_0^2\frac{\partial^2 V}{\partial x^2} + \left(r_0 - D_0 + \frac{3-5\alpha}{4}\overline{\sigma}_0^2\right)\frac{\partial V}{\partial x} \\ &+ \left(\frac{1-\alpha}{2}(r_0 - D_0) + \frac{(\alpha-1)(3\alpha-1)}{24}\overline{\sigma}_0^2 - (r_0 + \xi\eta_0)e^{-(1-\alpha)x}\right)V.\end{aligned}$$

The corresponding generalized Hamiltonian operator associated to this equation reads

$$\begin{aligned}\hat{H} =& \frac{1}{2}\overline{\sigma}_0^2\frac{\partial^2}{\partial x^2} + \left(r_0 - D_0 + \frac{3-5\alpha}{4}\overline{\sigma}_0^2\right) \\ &\frac{\partial}{\partial x} + \underbrace{\frac{1-\alpha}{2}(r_0 - D_0) + \frac{(\alpha-1)(3\alpha-1)}{24}\overline{\sigma}_0^2 - (r_0 + \xi\eta_0)e^{-(1-\alpha)x}}_{W(x)},\end{aligned}$$

where $W(x)$ is the effective potential of the theory. This Hamiltonian is comparable to the one obtained by Baaquie (2004) and Baaquie et al. (2004).

Qualitative and computational analyses

Provided that $V(S, 0) = \max(S - K, 0)$, where K is the strike price, the price of an up-and-out calloption in fractal dimension is then given by Lo and Hui (2011):

$$
\begin{aligned}
V(x,t) &= \int_{-\infty}^{0} dx' \left(\tilde{G}(x,t;x',0) - \tilde{G}(x,t;-x',0)e^{-\zeta x'}\right) V(x',0), \\
&= e^{S_0\left(x+(A_\alpha - P_\alpha)t + \frac{1}{\beta}\left(B_\alpha - Q_\alpha + \frac{1}{2}\overline{\sigma}_0^2\right)t^\beta\right)} \\
&\left(\mathbf{N}\left(-\frac{x+A_\alpha t + \frac{1}{\beta}\left(B_\alpha + \frac{1}{2}\overline{\sigma}_0^2\right)t^\beta}{\sqrt{\frac{1}{\beta}\overline{\sigma}_0^2 t^\beta}}\right) - \mathbf{N}\left(-\frac{x+A_\alpha t + \frac{1}{\beta}\left(B_\alpha + \frac{1}{2}\overline{\sigma}_0^2\right)t^\beta - \ln\left(\frac{K}{S_0}\right)}{\sqrt{\frac{1}{\beta}\overline{\sigma}_0^2 t^\beta}}\right)\right) \\
&- Ke^{-P_\alpha t - \frac{Q_\alpha}{\beta}t^\beta}\left(\mathbf{N}\left(-\frac{x+A_\alpha t + \frac{B_\alpha}{\beta}t^\beta}{\sqrt{\frac{1}{\beta}\overline{\sigma}_0^2 t^\beta}}\right) - \mathbf{N}\left(-\frac{x+A_\alpha t + \frac{B_\alpha}{\beta}t^\beta - \ln\left(\frac{K}{S_0}\right)}{\sqrt{\frac{1}{\beta}\overline{\sigma}_0^2 t^\beta}}\right)\right) \\
&- e^{S_0\left(-P_\alpha t - \frac{Q_\alpha}{\beta}t^\beta + (\zeta-1)\left(x+A_\alpha t + \frac{B_\alpha}{\beta}t^\beta\right) + \frac{1}{2\beta}(\zeta-1)^2\overline{\sigma}_0^2 t^\beta\right)} \\
&\times \left(\mathbf{N}\left(\frac{x+A_\alpha t + \frac{1}{\beta}(B_\alpha + (\zeta-1)\overline{\sigma}_0^2)t^\beta}{\sqrt{\frac{1}{\beta}\overline{\sigma}_0^2 t^\beta}}\right) - \mathbf{N}\left(\frac{x+A_\alpha t + \frac{1}{\beta}(B_\alpha + (\zeta-1)\overline{\sigma}_0^2)t^\beta + \ln\left(\frac{K}{S_0}\right)}{\sqrt{\frac{1}{\beta}\overline{\sigma}_0^2 t^\beta}}\right)\right) \\
&+ Ke^{-P_\alpha t - \frac{Q_\alpha}{\beta}t^\beta + \zeta\left(x+A_\alpha t + \frac{B_\alpha}{\beta}t^\beta\right) + \frac{1}{2\beta}\zeta^2\overline{\sigma}_0^2 t^\beta} \\
&\times \left(\mathbf{N}\left(\frac{x+A_\alpha t + \frac{1}{\beta}(B_\alpha + \overline{\sigma}_0^2)t^\beta}{\sqrt{\frac{1}{\beta}\overline{\sigma}_0^2 t^\beta}}\right) - \mathbf{N}\left(\frac{x+A_\alpha t + \frac{1}{\beta}(B_\alpha + \overline{\sigma}_0^2)t^\beta + \ln\left(\frac{K}{S_0}\right)}{\sqrt{\frac{1}{\beta}\overline{\sigma}_0^2 t^\beta}}\right)\right),
\end{aligned}
\tag{25}
$$

where ζ is an adjustable parameter controlling the barrier's movement and assumed to be

$$x^*(t) = \ln\left(\frac{S^*}{S_0}\right) = -A_\alpha t + \frac{2B_\alpha - \zeta\overline{\sigma}_0^2}{2\beta}t^\beta. \tag{26}$$

Here $\mathbf{N}(\lambda) = \int_{-\infty}^{\lambda} e^{-s^2/2}ds \Big/ \sqrt{2\pi} = \left(erf\left(\lambda \Big/ \sqrt{2}\right) + 1\right)\Big/ 2$ is the cumulative distribution function for the standard normal distribution expressed in terms of the error function $erf(z) = 2\int_0^z e^{-t^2}dt \Big/ \sqrt{\kappa}$. The optimal value of the controlling parameter ζ is given by

$$\zeta_{opt} = (1-2\beta)\overline{\sigma}_0^{-2}\frac{\beta A_\alpha \tau^{2-\beta} + B_\alpha \tau}{2\beta^2}, \tag{27}$$

where τ is the time at which the option price is evaluated. Notably, the upper (lower) bound for the exact barrier option prices can be obtained by the option price correlated with a moving barrier, such that $x^*(t)$ is the non-negative (non-positive) function for the duration of interest (Lo and Hui 2011).

To estimate α and β, we select $K = 11$, $S_0 = 15$, $S = 10$, $x = \ln 10/15 = -0.405$, $\overline{\sigma}_0^2 = 0.2$, $r_0 = 0.1$, $D_0 = 0.2$, $t = \tau = 1$ and $t_0 = 0.1$. If, first, we fix $\alpha = 1$, we can evaluate the value of β for which $V(1,1) = 0.509009$ obtained in Lo and Hui (2011) and Lo et al. (2003) for the case of pricing-barrier options. We find, using MATLAB code, the numerical estimate $\beta \approx 0.9471$. In contrast, if we fix $\beta = 1$, we obtain $\alpha \approx 0.9831$. One may verify that by changing the numerical values of the theory's parameters, the ranges of α and β are both in the limit $(0.9, 1)$. In fact, Vedder (1987) observes that $erf(x) \approx tanh(\overline{a}x + \overline{b}x^3)$ for $\overline{a} = 123/109$ and $\overline{b} = 11/109$ with a magnitude of difference $tanh(\overline{a}x + \overline{b}x^3) - erf(x) < 3.61 \times 10^{-4}$. Selecting, for numerical illustration purpose, the following numerical values $\overline{\sigma}_0^2 = 2$, $S_0 = 1$, $K = 1/2$, $t_0 = 0.1$, $r_0 \approx D_0$,and $\alpha = \beta = 0.97$, we find $\zeta_{opt} \approx 0.16$, $P_\alpha = \overline{P}_\alpha \approx 0$, $Q_\alpha = 0.0316$, $\overline{Q}_\alpha = 0.321$, and

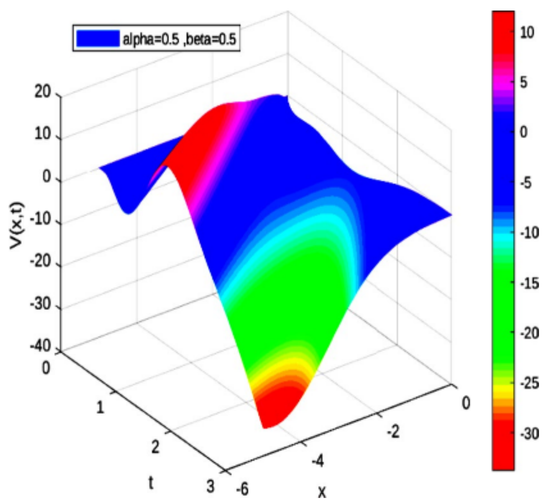


Fig. 1 Plot $V = V(x, t, 0.5, 0.5)$

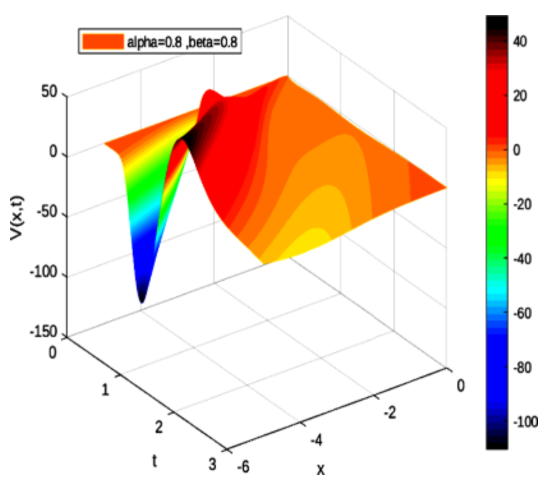


Fig. 2 Plot $V = V(x, t, 0.8, 0.8)$

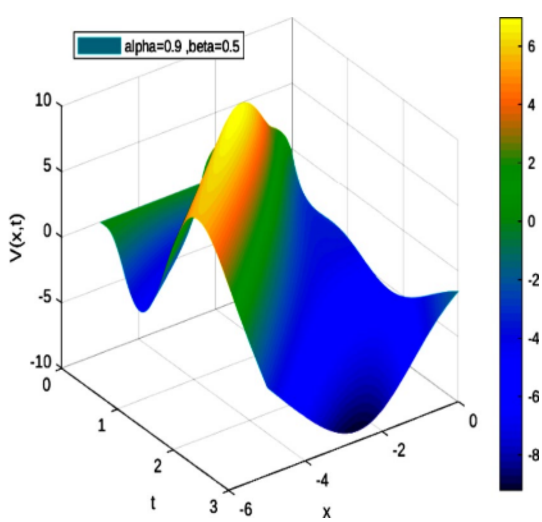


Fig. 3 Plot $V = V(x, t, 0.9, 0.5)$

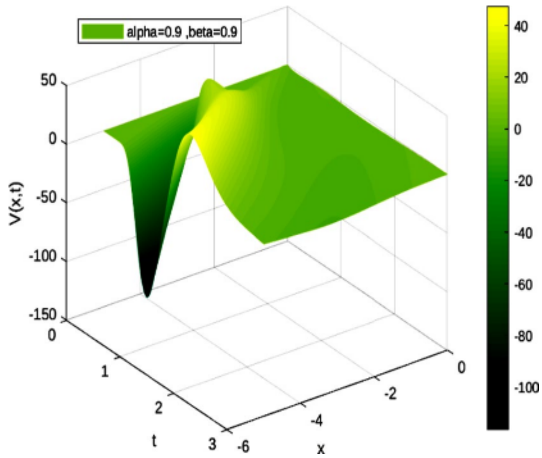


Fig. 4 Plot $V = V(x, t, 0.9, 0.9)$

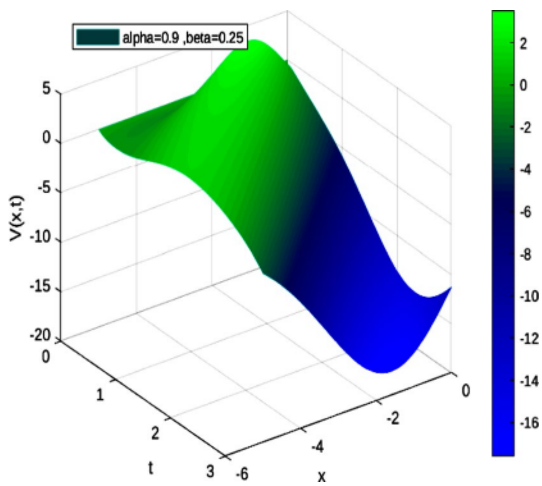


Fig. 5 Plot $V = V(x, t, 0.9, 0.25)$

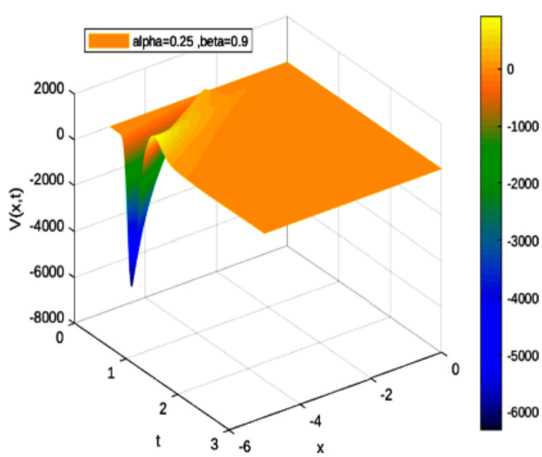


Fig. 6 Plot $V = V(x, t, 0.25, 0.25)$

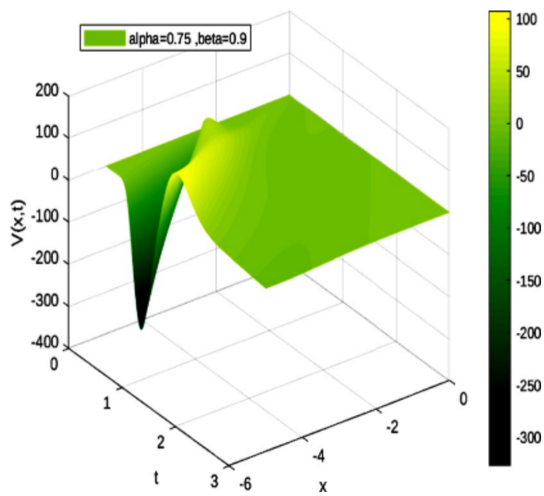


Fig. 7 Plot $V = V(x, t, 0.75, 0.9)$

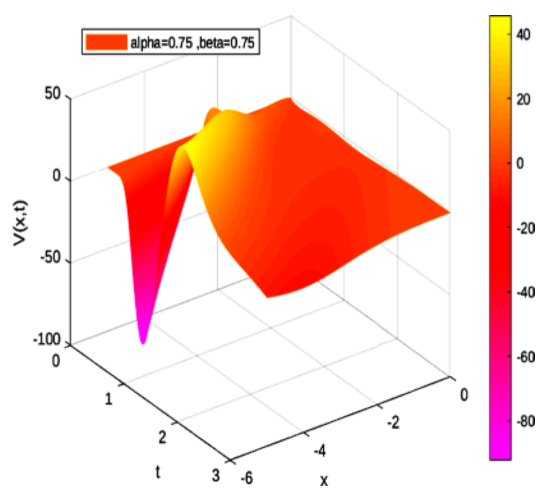


Fig. 8 Plot $V = V(x, t, 0.75, 0.75)$

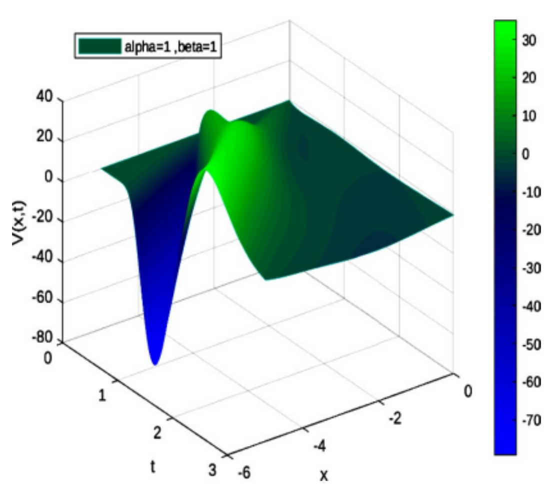


Fig. 9 Plot $V = V(x, t, 1, 1)$

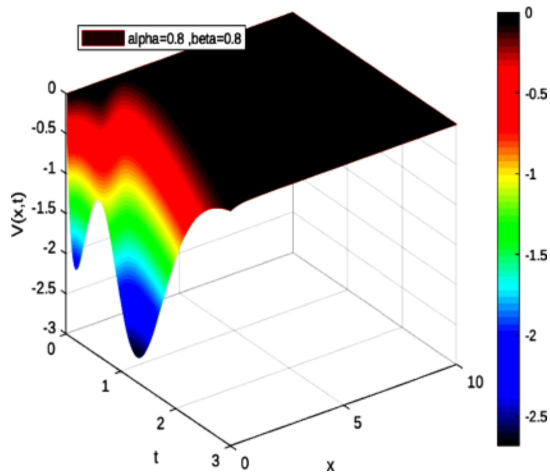


Fig. 10 Plot $V = V(x, t, 0.8, 0.8)$

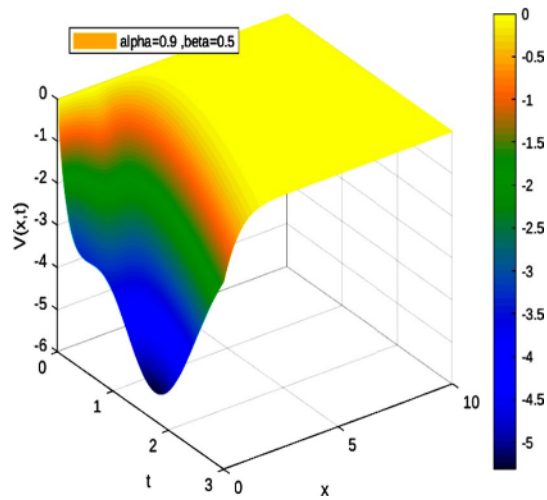


Fig. 11 Plot $V = V(x, t, 0.9, 0.5)$

$B_\alpha = -0.646$. Some numerical estimates are as follow (obtained using the software packages Wolfram Alpha and Maple):

- for $K = S = 1$, the solution $V(-0.6, 1, \alpha, 1) = -0.509009$ does not have any solution, whereas for $K = 3$ and $S = 1$, we find $\alpha = 0.89114$,
- for $K = 4$ and $S = 1$, the solution $V(-0.6, 1, \alpha, 1) = -0.509009$ gives $\alpha = 0.938488$ whereas for $K = 5$ and $S = 1$, we obtain $\alpha = 0.9546160$,
- for $K = S = 1$, the solution $V(-0.6, 1, 1, \beta) = -0.509009$ does not have any solution, whereas for $K = 2, 3, 4$ and $S = 1$, the solution returns a complex solution, and for $K = 5$ and $S = 1$, we get $\beta = 0.65399979$.

We plot the following numerical solutions (Figs. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31 and 32): $V = V(x, t, \alpha, \beta)$

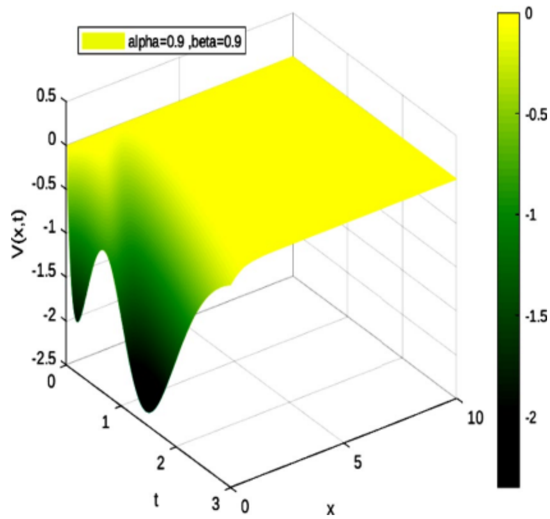


Fig. 12 Plot $V = V(x, t, 0.9, 0.9)$

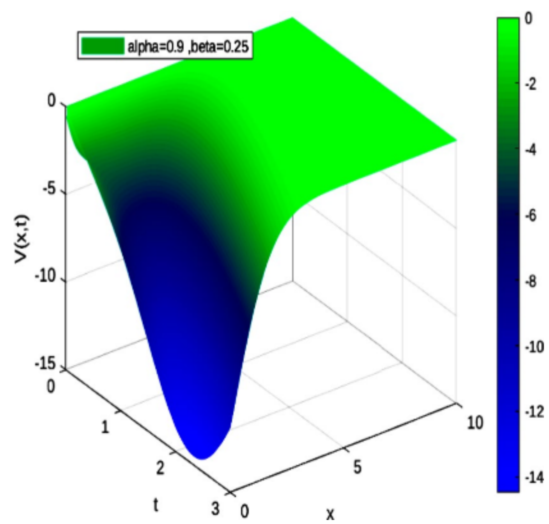


Fig. 13 Plot $V = V(x, t, 0.9, 0.25)$

for $K = 11, S_0 = 1, \overline{\sigma}_0^2 = 1$ and both for different values of α and β, $t_0 = 0.1$,$r_0 \approx D_0$ and various scales:

Discussion and robustness analysis

The principal features are that $V(x, t)$ is affected by fractal dimensions, and the variations of $V(x, t)$ have major modifications. Our numerical results show that, for fractal dimensions $\alpha \in (0.9, 1)$ and $\beta \in (0.9, 1)$, the value of the asset changes in a plausible way compared to Figs. 6, 14, 15, 19, and 29,which correspond to fractal dimensions much less than unity, since the asset is close to being at-the-money near expiry. Additionally, the associated value changes dramatically as $V(x, t)$ crosses the strike price. The

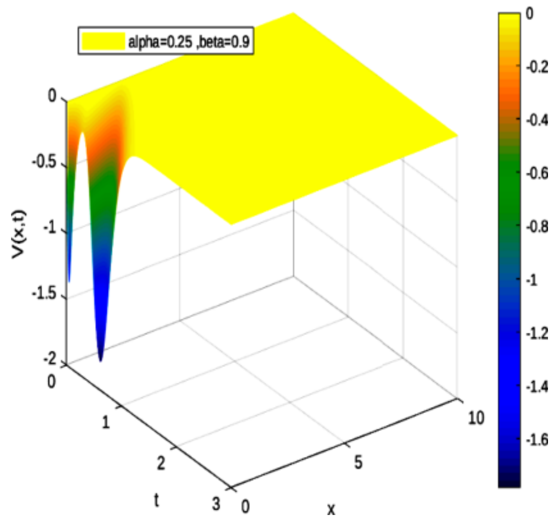


Fig. 14 Plot $V = V(x, t, 0.25, 0.9)$

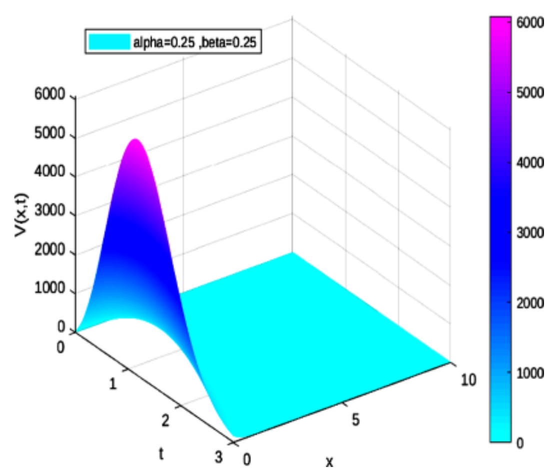


Fig. 15 Plot $V = V(x, t, 0.25, 0.25)$

option-pricing model tends to misprice out-of-the-money options at a larger extent than in-the-money options for fractal dimensions less $0 < \alpha < 0.5$ and $0 < \beta < 0.5$. This is expected, since we need to have fractal dimensions close to unity for our model to be in accordance with the outcomes found in the literature. The results, as shown in these figures, indicate that the fractal dimension has a significant effect on dividend payout and hence on the option price and stocks markets. We also observe that the emergence of fluctuations or abnormal behavior mainly when the fractal dimension is much smaller than unity is due to the nature of the solution obtained. This is due to two main effects: the effect of a variable's volatility on option prices and the fractal dimension effect. Volatility, in principle, refers to the fluctuations in the underlying asset's market price. Therefore, its variations with both current stock price and time may be used to spot patterns in market behavior (Peters 1994).

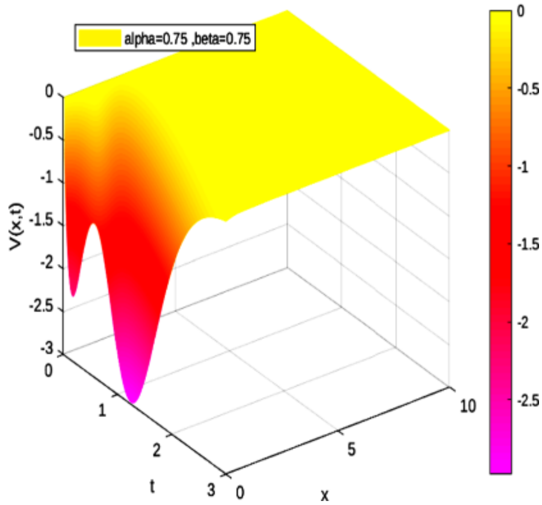


Fig. 16 Plot $V = V(x, t, 0.75, 0.75)$

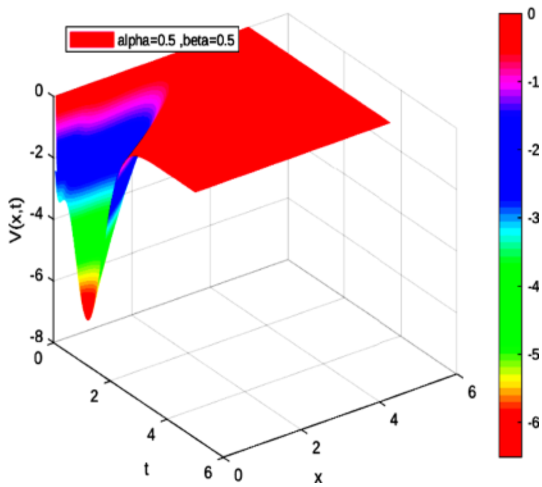


Fig. 17 Plot $V = V(x, t, 0.5, 0.5)$

Verifying that that the option values increase considerably when the stock prices increase and when the fractal dimensions approach unity is easy. It should be stressed that these solutions strongly depend on the numerical choices of the free parameters in the theory and on the scales selected. To show this, we select the case where $\zeta = 2$. Some graphs (Figs. 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, and 48) are plotted below for different choices of fractal dimensions:

These graphs illustrate how fractal dimensions influence the dividend payout of the BSE for a wider range of scales. One observes a difference of shapes for different scales and different upper levels of V. Recall that the solutions in terms of the current stock price and time—$V(S, t)$—differ from $V(x, t)$, as shown in Figs. 49, 50, 51, 52, 53 and 54:

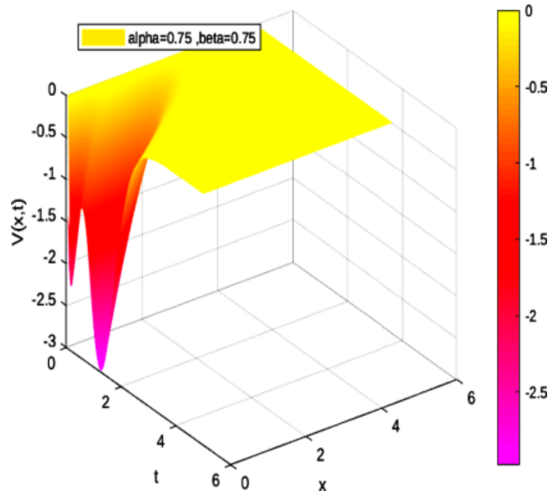


Fig. 18 Plot $V = V(x, t, 0.75, 0.75)$

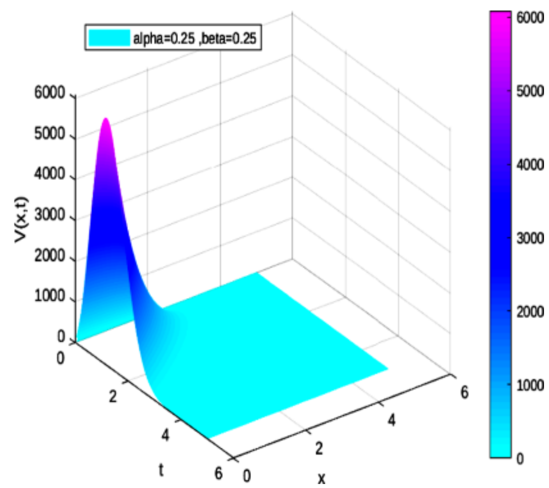


Fig. 19 Plot $V = V(x, t, 0.25, 0.25)$

These graphical solutions of call option price obtained from the two-dimensional BS model in fractal dimensions show that, by decreasing the numerical values of the fractal dimensions, the call option increases significantly. For fixed time, the option's price increases as the stock price increases, and for fixed S and decreasing time (i.e., approaching maturity), the call becomes negligible, since its value at expiration is assured. In fact, an increase in the underlying asset's price leads to an increase in the value of the call option, and hence, the probability that the option will be exercised is high. Nevertheless, high call options could be risky owing to the potential for devastating losses for traders if the stock soars.In all cases, the general behaviors of these solutions are in good agreement with the findings obtained in the literature (Edeki et al. 2015; Khalsaraei et al. 2022) and even those based on fractional derivatives (Golbabai et al. 2019; Ravi Kanth and Aruna 2016; Saratha et al. 2020). Hence, our fractal

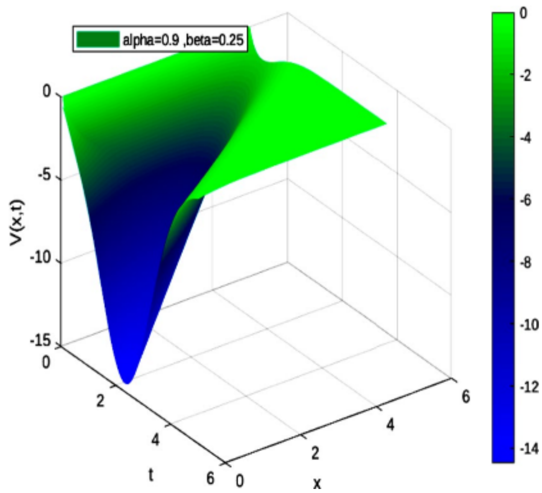


Fig. 20 Plot $V = V(x, t, 0.9, 0.25)$

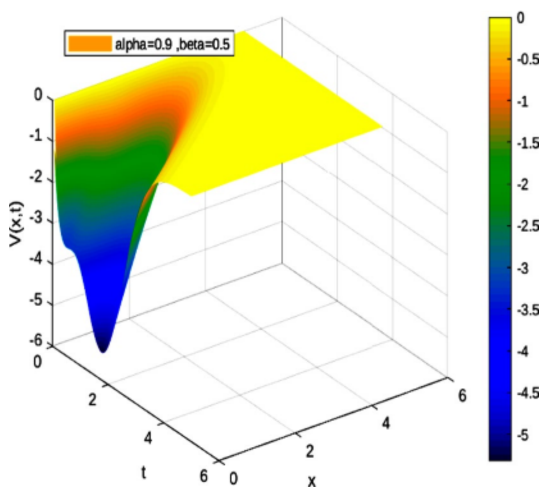


Fig. 21 Plot $V = V(x, t, 0.9, 0.5)$

approach also gives comparable results to those studies based on fractional derivatives characterized by the memory-less property.

Understanding how fractals dimensions misprice the options prices is a tricky problem. The analysis of the BSE's behavior in fractal dimensions and in the presence of variable parameters may offer an answer to this problem to a certain extent. More specific and intangible factors are also required to obtain a plausible solution, although we believe that no mathematical model will ever be able to predict an option's exact value. Recall that these results hold if the following power-laws apply:$\overline{\sigma}^2(t) = \overline{\sigma}_0^2 t^{\beta-1}$, $r(S,t) = r_0 t^{1-\beta} S^{\alpha-1}$,and $D(S,t) = D_0 t^{1-\beta} S^{\alpha-1}$. In fact, the impacts of interest rate on stock exchange offer substantial implications for economic growth and control of a nation's overall money (monitory policy) and government policy toward financial markets (Alam and Salah Uddin 2009; Doong et al. 2005; Gu et al. 2021; McMillan 2022). Notably, for $D(t) = 0$, the solution of Eq. (15) may be

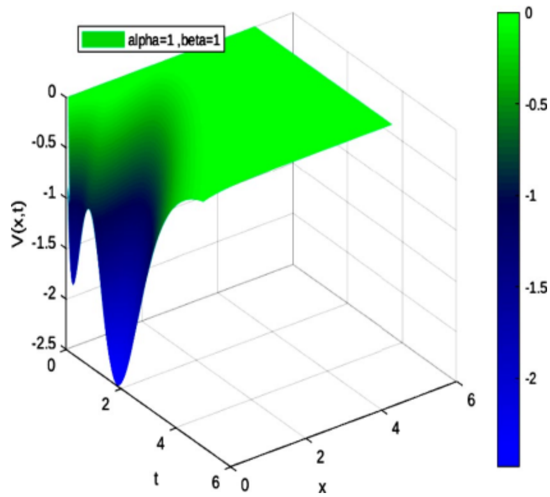


Fig. 22 Plot $V = V(x, t, 1, 1)$

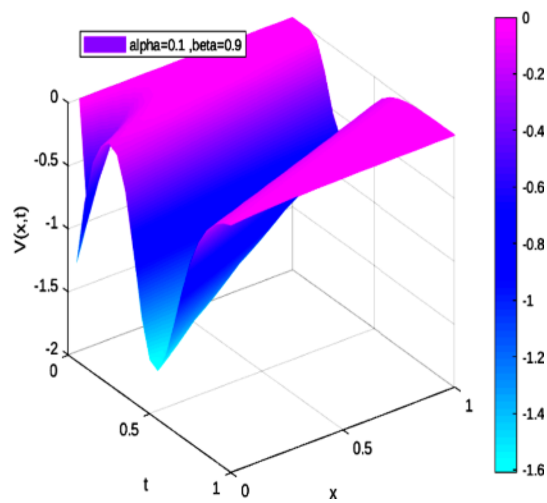


Fig. 23 Plot $V = V(x, t, 0.1, 0.9)$

obtained without using the method of image but with the aid of general transformations introduced by Ghevariya (2019). The price of a European call option on a non-dividend-paying equity is, in fact, split into three terms consisting of a Black–Scholes price for the constant-coefficient case in a non-dividend-paying set-up, the ratio of two strike prices $K \big/ \tilde{K}$, and a parameterized time $\tilde{t}$. It is given, after algebra, by

$$V(S,t) = \frac{K}{\tilde{K}} e^{-\int_t^T \left(r(s) - \frac{r}{\tilde{\sigma}^2} \overline{\sigma}^2(s) \right) ds} \tilde{V}(\tilde{S}, \tilde{t}), \tag{28}$$

where $\tilde{V}(\tilde{S}, \tilde{t})$ (value of call option having asset price $\tilde{S}$ at time $\tilde{t} < \tilde{T}$) is the solution of the subsequent BSE with constant parameters:

$$\frac{\partial \tilde{V}}{\partial \tilde{t}} = \frac{1}{2} \tilde{\sigma}^2 \tilde{S}^2 \frac{\partial^2 \tilde{V}}{\partial \tilde{S}^2} + \tilde{r} \tilde{S} \frac{\partial \tilde{V}}{\partial \tilde{S}} - \tilde{r} \tilde{V}. \tag{29}$$

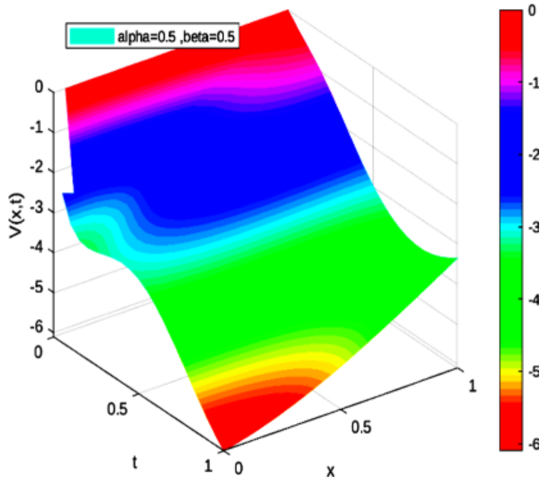


Fig. 24 Plot $V = V(x, t, 0.5, 0.5)$

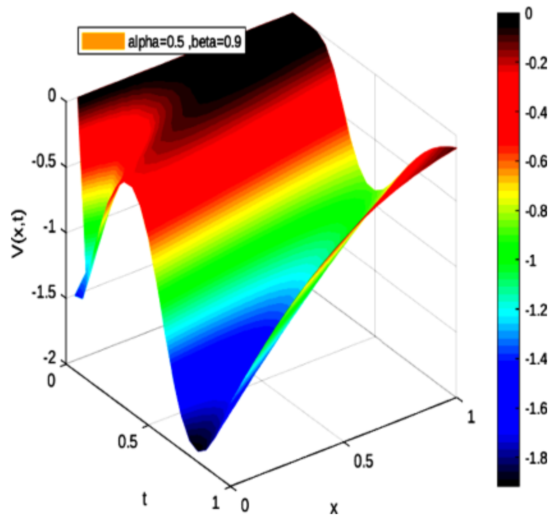


Fig. 25 Plot $V = V(x, t, 0.5, 0.9)$

This equation is characterized by the payoff $V(\tilde{S},0) = \max(\tilde{S}^p - \tilde{K}, 0), p > 0$ for the standard powered call option. Here, $\tilde{\sigma}^2$ and $\tilde{r}$ are the constant volatility and risk-free interest rate, respectively. Additionally, we have

$$\tilde{S} = \left(\frac{\tilde{K}}{K}\right)^{\frac{1}{p}} e^{-\int_t^T \left(\frac{\tilde{r}}{\tilde{\sigma}^2}\overline{\sigma}^2(s) - \overline{r}(s)\right)ds} = \left(\frac{\tilde{K}}{K}\right)^{\frac{1}{p}} e^{-\left(\frac{\tilde{r}}{\tilde{\sigma}^2}\overline{\sigma}_0^2 - Q_\alpha\right)\frac{1}{\beta}\left(T^\beta - t^\beta\right) + P_\alpha(T-t)}, \tag{30}$$

$$\tilde{t} = -\frac{1}{\tilde{\sigma}^2}\int_t^T \overline{\sigma}^2(s)ds + \tilde{T} = -\frac{\overline{\sigma}_0^2}{\tilde{\sigma}^2}\frac{1}{\beta}\left(T^\beta - t^\beta\right) + \tilde{T}, \tag{31}$$

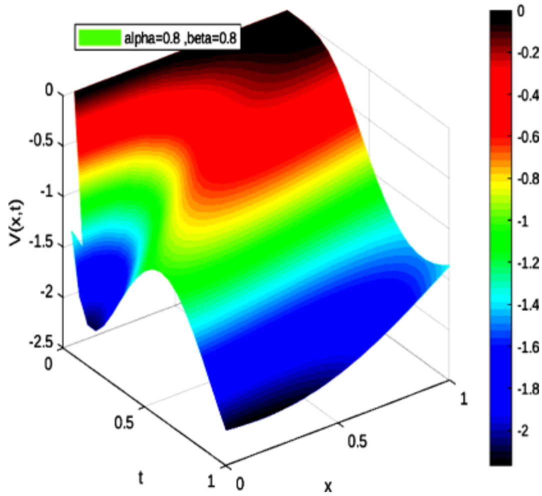


Fig. 26 Plot $V = V(x, t, 0.8, 0.8)$

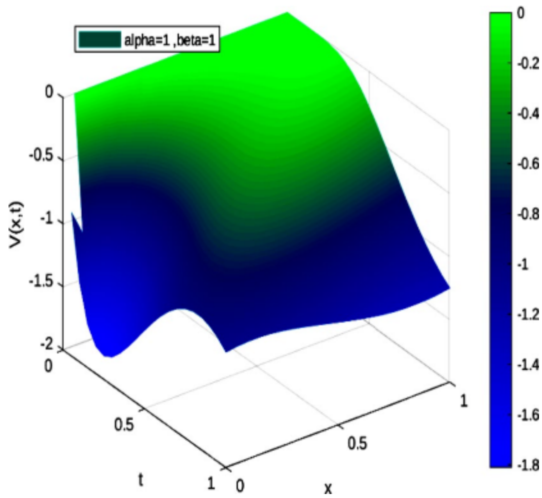


Fig. 27 Plot $V = V(x, t, 1, 1)$

$$\begin{aligned} \tilde{V}(\tilde{S},\tilde{t}) &= \tilde{S}^p e^{(p-1)\left(\tilde{r}+\frac{p}{2}\tilde{\sigma}^2\right)\left(\tilde{T}-\tilde{t}\right)} N(d_1) - \tilde{K}e^{-\tilde{r}\left(\tilde{T}-\tilde{t}\right)} N(d_2), \\ &= \frac{\tilde{K}}{K} e^{-\left(\frac{\tilde{r}}{\tilde{\sigma}^2}\overline{\sigma}_0^2 - Q_\alpha\right)\frac{p}{\beta}\left(T^\beta - t^\beta\right) + pP_\alpha(T-t)} e^{\left(\tilde{r}+\frac{p}{2}\tilde{\sigma}^2\right)\frac{\overline{\sigma}_0^2}{\tilde{\sigma}^2}\frac{p-1}{\beta}\left(T^\beta - t^\beta\right)} N(d_1) - \tilde{K}e^{-\frac{\overline{\sigma}_0^2}{\tilde{\sigma}^2}\frac{\tilde{r}}{\beta}\left(T^\beta - t^\beta\right)} N(d_2), \end{aligned} \tag{32}$$

with

$$= \frac{\ln\left(\frac{1}{K}\left(\frac{\tilde{K}}{K}\right)^{\frac{1}{p}}\right) + P_\alpha(T-t) + \left((2p-1)\frac{\bar{\sigma}_0^2}{2} + Q_\alpha\right)\frac{1}{\beta}\left(T^\beta - t^\beta\right)}{\bar{\sigma}_0\sqrt{\frac{1}{\beta}\left(T^\beta - t^\beta\right)}} \tag{33}$$

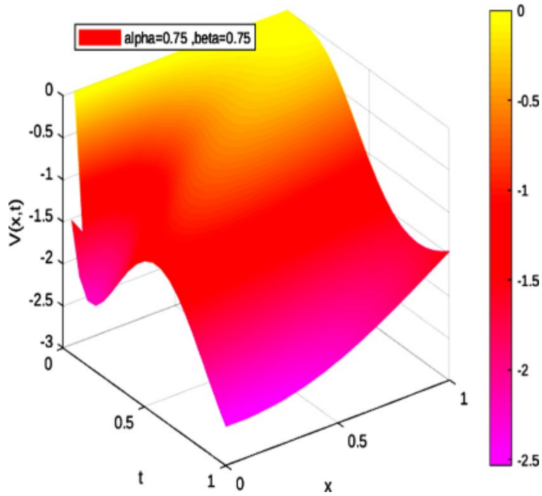


Fig. 28 Plot $V = V(x, t, 0.75, 0.75)$

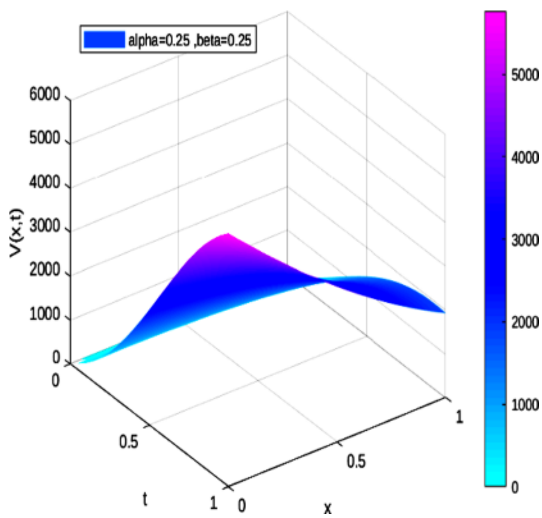


Fig. 29 Plot $V = V(x, t, 0.25, 0.25)$

$$
\begin{aligned}
d_2 &= \frac{\ln\left(\frac{\tilde{S}}{\tilde{K}}\right) + \left(\tilde{r} - \frac{\tilde{\sigma}^2}{2}\right)\left(\tilde{T} - \tilde{t}\right)}{\tilde{\sigma}\sqrt{\tilde{T} - \tilde{t}}}, \\
&= \frac{\ln\left(\frac{1}{\tilde{K}}\left(\frac{\tilde{K}}{K}\right)^{\frac{1}{p}} e^{-\left(\frac{\tilde{r}}{\tilde{\sigma}^2}\overline{\sigma}_0^2 - Q_\alpha\right)\frac{1}{\beta}\left(T^\beta - t^\beta\right) + P_\alpha(T-t)}\right) + \left(\tilde{r} - \frac{\tilde{\sigma}^2}{2}\right)\frac{\overline{\sigma}_0^2}{\tilde{\sigma}^2}\frac{1}{\beta}\left(T^\beta - t^\beta\right)}{\overline{\sigma}_0\sqrt{\frac{1}{\beta}\left(T^\beta - t^\beta\right)}}, \\
&= \frac{\ln\left(\frac{1}{\tilde{K}}\left(\frac{\tilde{K}}{K}\right)^{\frac{1}{p}}\right) + P_\alpha(T-t) + \left(Q_\alpha - \frac{\overline{\sigma}_0^2}{2}\right)\frac{1}{\beta}\left(T^\beta - t^\beta\right)}{\overline{\sigma}_0\sqrt{\frac{1}{\beta}\left(T^\beta - t^\beta\right)}}.
\end{aligned}
\tag{34}
$$

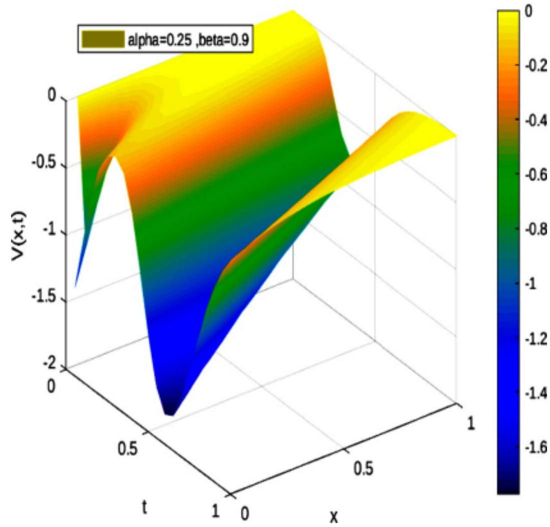


Fig. 30 Plot $V = V(x, t, 0.75, 0.75)$

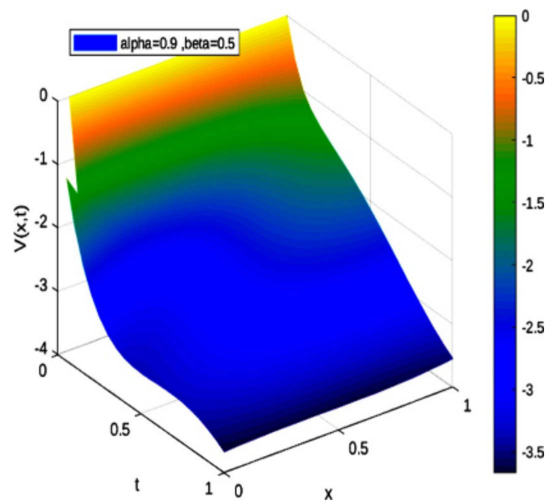


Fig. 31 Plot $V = V(x, t, 0.9, 0.5)$

This result shows that the price of a European call option on a non-dividend-paying equity in fractal dimensions may also be obtained using this methodology. In some special cases, obtaining an asymptotic form of the BSE using financial reasoning and based on specific boundary conditions is rather straightforward—for instance, Dirichlet, Neumann, linear asymptotic boundary conditions and partial differential equation (PDE) boundary conditions. In fact,

- The Dirichlet boundary condition is practical for simple dynamic equations (Company et al. 2008; Hajipour and Malek 2015; Kangro and Nicolaides 2000); yet, it is tricky for complex numerical systems such as those used to solve the BSE as it depends on the payoff of the derivative contract being priced and fluctuations in the underlying asset, as well as on the nature of the stochastic process followed by

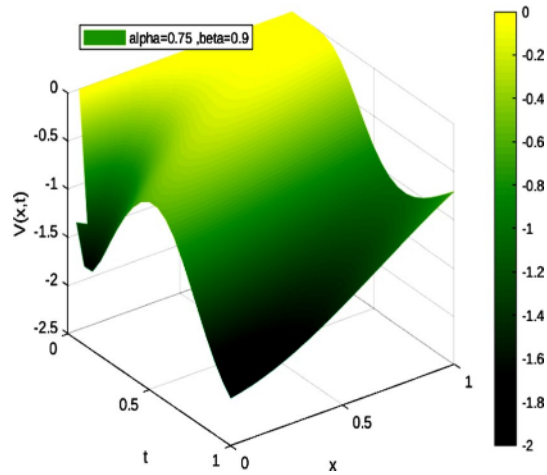


Fig. 32 Plot $V = V(x, t, 0.75, 0.9)$

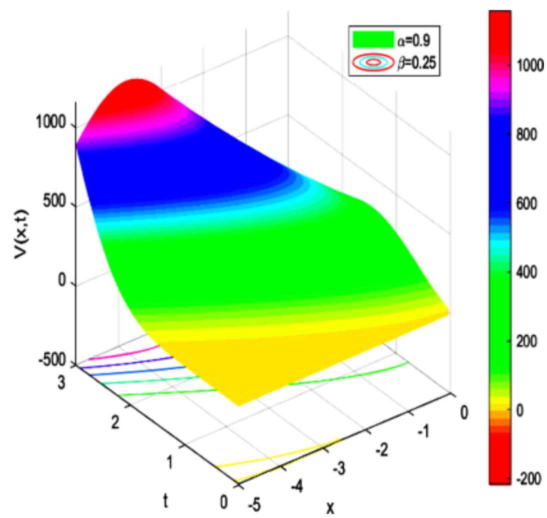


Fig. 33 Plot $V = V(x, t, 0.9, 0.25)$ for $\zeta = 2$

the underlying asset. Moreover, since prices can also influenced by risk premia and hence add more complication for the problem, path-dependent options with jump conditions add extra impediment.

- The Neumann boundary condition in its turn fixes numerical values for the derivative of the asymptotic solution at the boundary of the spatial domain (Kurpiel and Roncalli 1999).
- The linear asymptotic boundary condition is frequently used in practice, particularly for the case of a complex path-dependent option. It simply assumes that the second derivative of the value of derivative security vanishes as the asset price becomes large (Linde et al. 2009).
- Finally, in the PDE boundary condition, one-sided discretization at the boundary points is used, so that the values of the ghost points are not required (Tavella and Randall 2000).

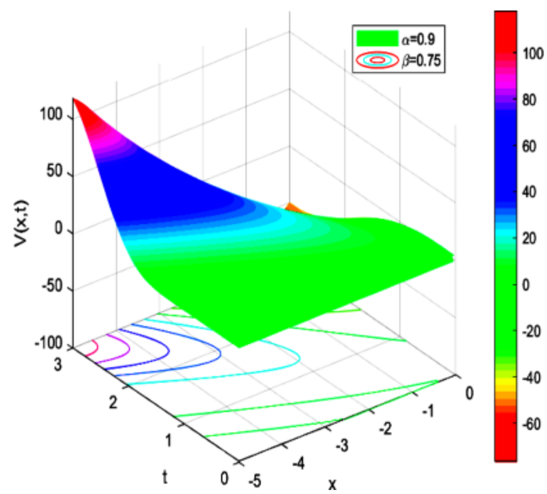


Fig. 34 Plot $V = V(x, t, 0.9, 0.75)$ for $\zeta = 2$

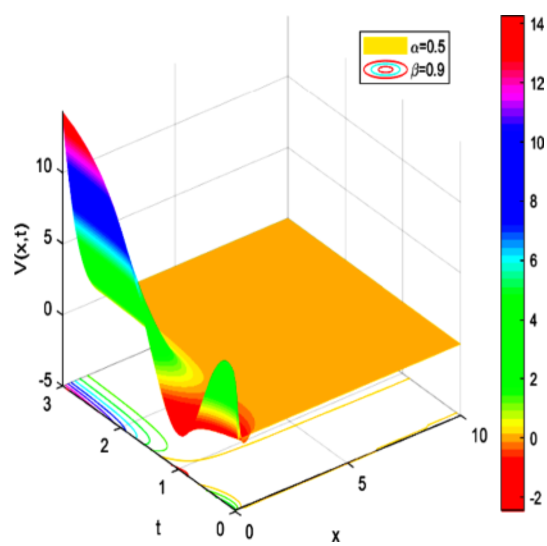


Fig. 35 Plot $V = V(x, t, 0.5, 0.9)$ for $\zeta = 2$

Fractional derivative models have *potential applications in analyzing economic growth*. In general, fractional calculus has been applied successfully to study stochastic volatility and stock pricing models, incorporating the Taylor formula and Brownian motion with drift and scaling. The literature includes several financial models based on fractional derivatives and fractal dimensions (Ampun et al 2022; Kachhia 2023; Li 2022; Liu et al 2022; Nuugulu et al 2022; Zhang et al 2022). Most of these are governed by fractional partial differential equations that necessitate advanced analytical and numerical techniques to model financial time-series and option prices—in particular, when the underlying price process is driven by a fractional Lévy process (Aljethi and Kilicman 2022). Some of these advanced analytical tools include new the homotopy transformation method (Zhang et al 2022) and the generalized Laplace variational iteration approach (Ampun et al 2022), among others (Diethelm et al 2022), in addition to the use

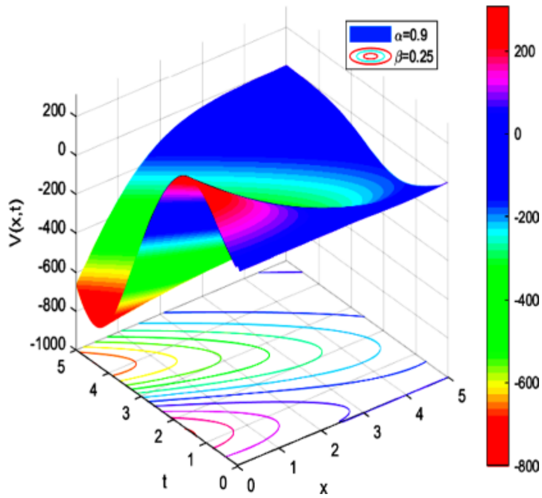


Fig. 36 Plot $V = V(x, t, 0.9, 0.25)$ for $\zeta = 2$

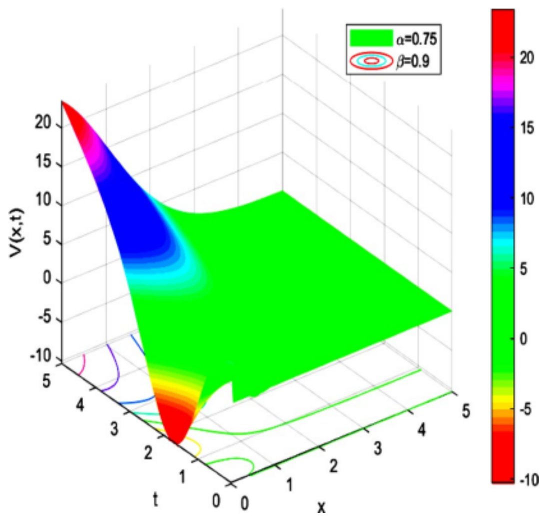


Fig. 37 Plot $V = V(x, t, 0.75, 0.9)$ for $\zeta = 2$

of efficient numerical methods (Gao et al 2023; Ma et al 2012; Rehman et al. 2024). Most of these numerical works neglect the analytic properties of the exact solutions to the partial differential equations, hence revealing ambiguous results. Our approach seems less demanding analytically. Moreover, the computational techniques and algorithmic and software tools employed in this study are based on Matlab and Mathematica, which renders our work less heavy than those studies based on fractional derivatives. Furthermore, our results are in agreement with recent studies (Edeki et al. 2015; Khalsaraei et al. 2022) and even those based on fractional calculus (Golbabai et al. 2019; Ravi Kanth and Aruna 2016; Saratha et al. 2020).

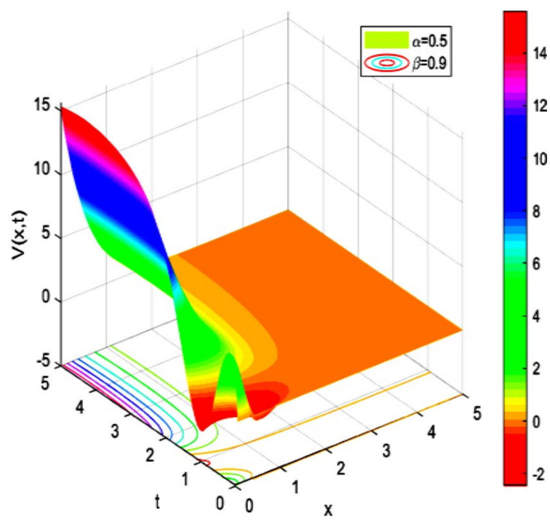


Fig. 38 Plot $V = V(x, t, 0.5, 0.9)$ for $\zeta = 2$

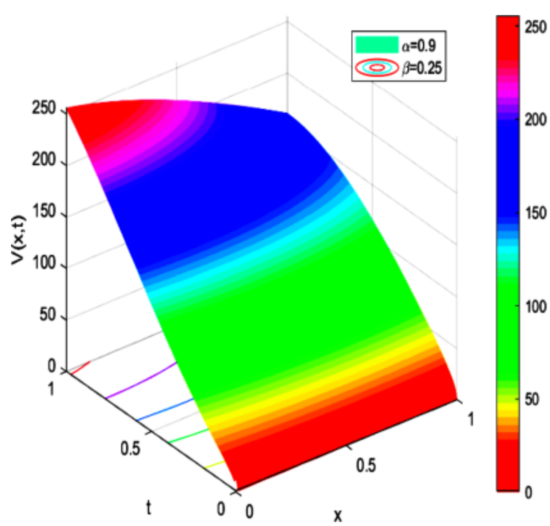


Fig. 39 Plot $V = V(x, t, 0.9, 0.25)$ for $\zeta = 2$

Conclusions

Jeong el al. (2018) recently presented a finite difference method for BSE without boundary conditions, which diminishes one or two computational grid points and hence estimates, at each time step, the novel solution on that new grid points. This tactic does not require the presence of boundary conditions, and it has proved practical for solving the BSE. One important point concerning the convergence and stability analysis of the present modelmust be mentioned at the end of this section. In the literature, several implicit finite difference methods have been considered for solving fractional BSE that are unconditionally stable (Ahmad and Nikan 2020; Nuugulu et al. 2021). For the numerical solution of generalized BSE, finite difference methods are considered as a potential numerical approach to estimate the price for European call and put options.

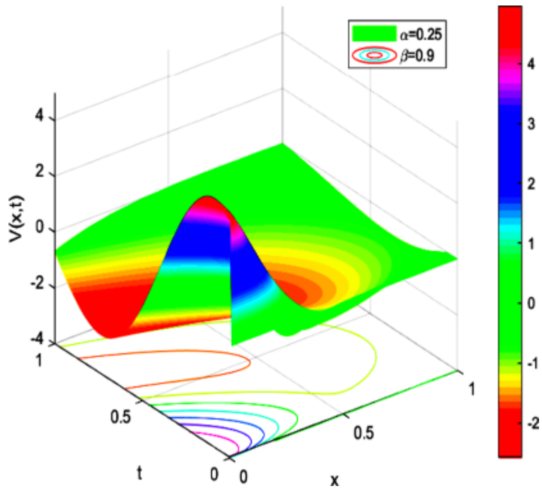


Fig. 40 Plot $V = V(x, t, 0.25, 0.9)$ for $\zeta = 2$

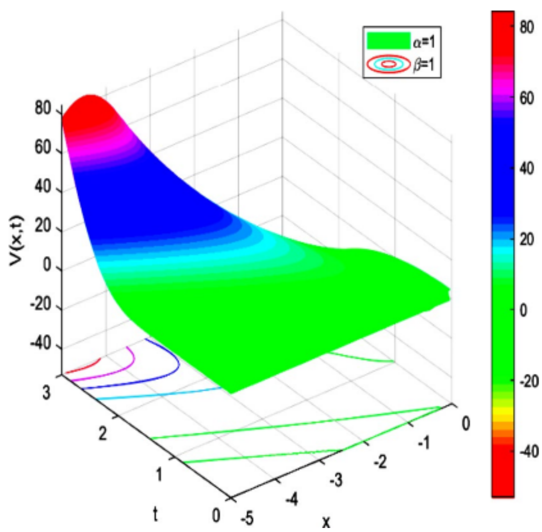


Fig. 41 Plot $V = V(x, t, 1, 1)$ for $\zeta = 2$

These numerical schemes can pass up probability theory and stochastic methods and effectively resolve the degeneracy. Additional numerical methods include Grunwald–Letnikov approximation (Scherer et al. 2011); the meshless local Petrov–Galerkin and implicit finite difference methods used for discretizing the governing equation in option price and time variable, respectively (Phaochoo and Luadsong 2015);the compact finite difference scheme for discretization of space derivative and described by means of Caputo and a compact finite difference method (Mesgarani et al. 2021; Roul and Goura 2021); and the posteriori grid method for solving the time-fractional BSE governing European options (Cen et al. 2022). Most of these numerical approaches support the convergence and stability of fractional BSE comparable to those obtained in the present study. The BSE in fractal markets has been solved using a package of numerical finite-difference method, as previously stated (Wang et al. 2022). Nevertheless, most studies

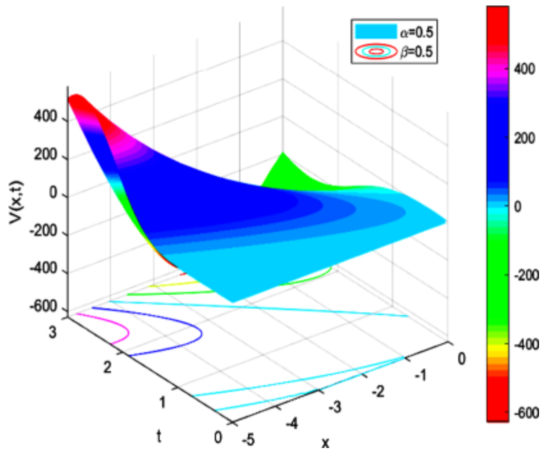


Fig. 42 Plot $V = V(x, t, 0.5, 0.5)$ for $\zeta = 2$

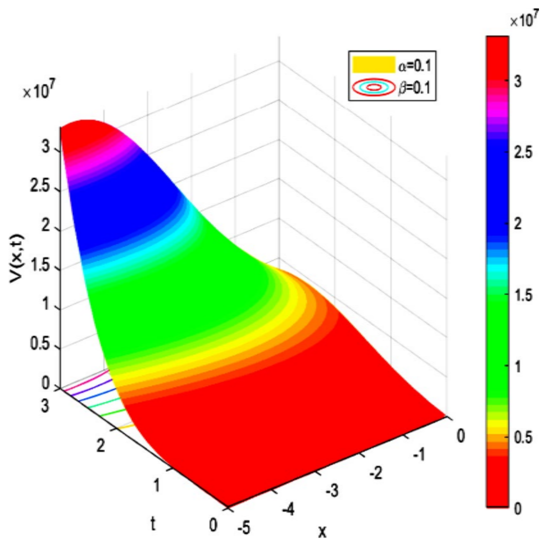


Fig. 43 Plot $V = V(x, t, 0.1, 0.1)$ for $\zeta = 2$

have focused on fractional Black–Scholes models for different options, such as the European put option (Prathumwan and Trachoo 2020). However, the construction of a fractional BSE for pricing European options—for example, on continuous dividend paying stocks—must be usually accompanied by a numerical package, such as the implicit finite difference method, owing to the difficulty in obtaining an analytical result (Nuugulu et al. 2021) or by developing new analytical methods to solve fractional BSE (Duan et al. 2018; Jena et al. 2020; Jumarie 2005, 2008, 2010b) in addition to those basic methodologies found in literature, such as the Laplace homotopy perturbation method (Kumar et al. 2012), the fractional variational iteration method (Elbeleze et al. 2013), the new Lagrange multipliers (Ghandehari and Ranjbar 2014), the homo-separation of variables (Ghandehari and Ranjbar 2016), the projected differential transformation method (Edeki et al. 2015), the multivariante Padé method (Özdemir and Yavuzb 2017), the meshless local Petrov Galerkin method (Phaochoo and Luadsong 2015; Phaochoo et al. 2016),

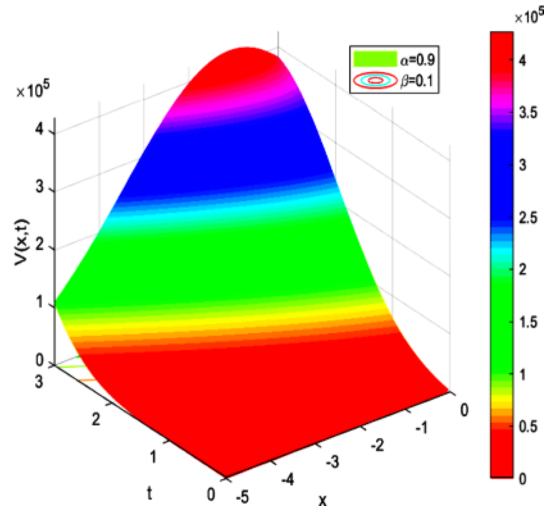


Fig. 44 Plot $V = V(x, t, 0.9, 0.1)$ for $\zeta = 2$

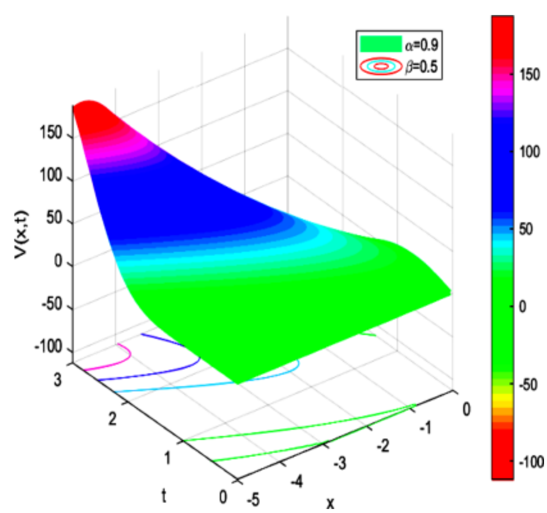


Fig. 45 Plot $V = V(x, t, 0.9, 0.5)$ for $\zeta = 2$

the fractional generalized homotopy analysis method (Saratha et al. 2020), the Sumudu transform and its derivatives (Khan and Ansari 2016), and the wavelet technique (Nourian et al. 2022).Review methods to solve the fractional BSE were recently proposed by Nurazizah et al. (2015). In this paper, we showed how a new fractal dimensional analysis of BSE could be used to study several important aspects in qualitative finance characterized by variables parameters. Our approach has not been based on fractional derivatives aspects that require a high-order computational technique for fractional BSE model describing option pricing (Roul 2022) but on fractal dimensional arguments. Notably, fractal dimensions are used for a wide range of random and stochastic processes, including fractional Brownian motion. Therefore, in future work, it will be motivating to discuss several financial options and compare the fractal dimension for different stocks markets, as well as to study their associated fractal BSE through the fractal dimension

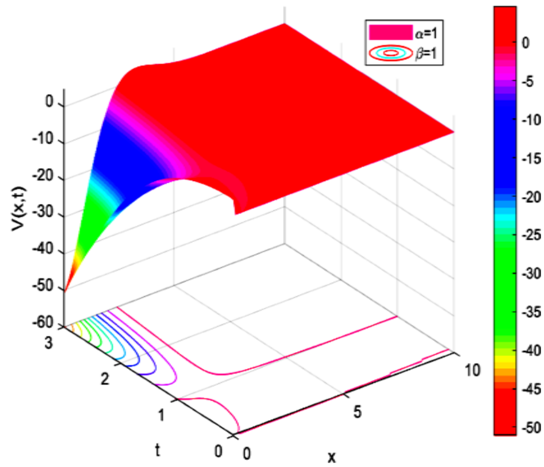


Fig. 46 Plot $V = V(x, t, 1, 1)$ for $\zeta = 2$

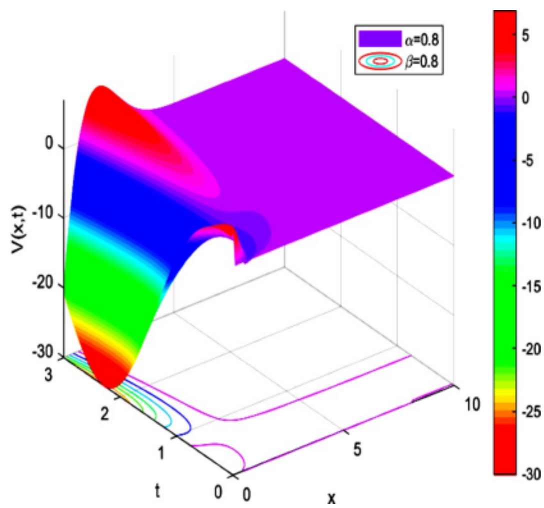


Fig. 47 Plot $V = V(x, t, 0.8, 0.8)$ for $\zeta = 2$

point of view. We believe that these studies will justify the use of the fractal dimension in qualitative finance. We point out that fractal calculus, fractal geometry, and fractional analysis are correlated to some extent, and the literature has highlighted a relationship between these two approaches (Butera and Di Paola 2014; Tatom 1995).

To conclude, the Black–Scholes model plays a leading role in quantitative finance owing to its truthful and practical estimation of stock prices. In this study, we have constructed a generalized BSE in fractal dimensions based on the concept of fractal derivatives introduced in fractal calculus. In fact, many studies have been devoted to study fractional processes in finance and economics and analyze their stability and convergence. However, the fractal approach is also correlated with economics and finance, and its implications in finance are highly appreciated. The fractal nature of several financial processes characterized by the anomalous distribution of market returns allows to model inefficient markets more accurately through the option-pricing model. In this work, we introduced a nonlinear generalization of the conventional BSE in which the

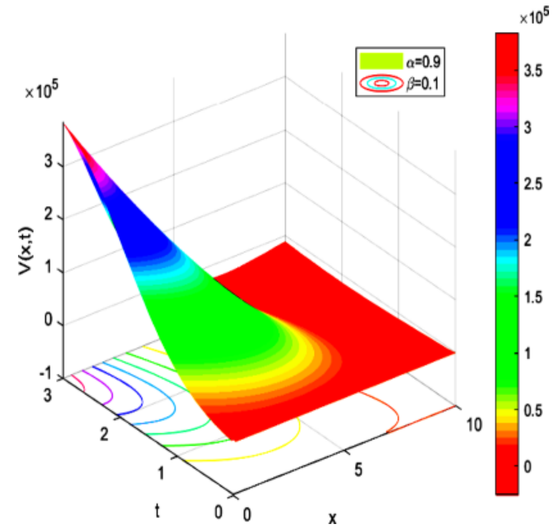


Fig. 48 Plot $V = V(x, t, 0.9, 0.1)$ for $\zeta = 2$

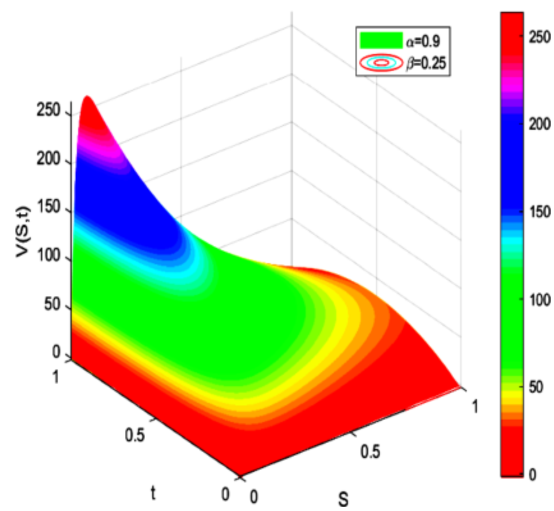


Fig. 49 Plot $V = V(S, t, 0.9, 0.25)$ for $\zeta = 2$

volatility, interest rate, and dividend payout are variables and obey fixed power-laws. Accordingly, we obtained a family of extended BSE that have proved to hold a number of motivating features in quantitative finance. For the pricing-barrier option, we evaluated the numerical estimates of the fractal dimensions and observed that α and β are within the range $(0.9, 1)$ and coincide with the recent outcomes found in the literature. The stock price is affected by fractal dimensions, and for the fractal dimensions $\alpha \in (0.9, 1)$ and $\beta \in (0.9, 1)$, the value of the asset changes slightly, whereas for $\alpha = \beta = 1$, the asset is close to being at-the-money near expiry. The price of a European call option on a non-dividend-paying equity in fractal dimensions may be obtained using the power-laws introduced in this study (Appleby et al. 2012; Bayraktar et al. 2004). The graphical solutions of call option price obtained from the two-dimensional BS model in fractal dimensions have shown that by decreasing the numerical values of the fractal dimensions, the

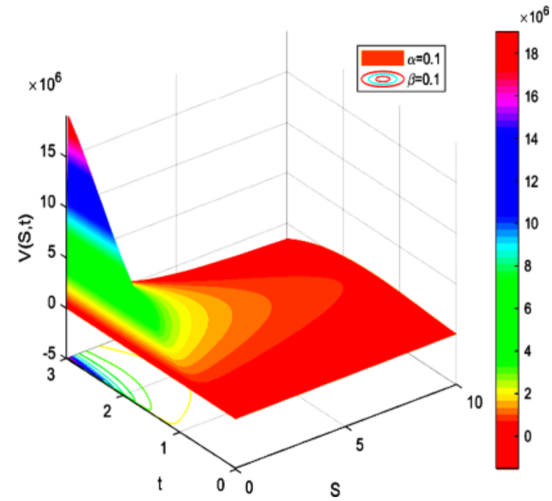


Fig. 50 Plot $V = V(S, t, 0.1, 0.1)$ for $\zeta = 2$

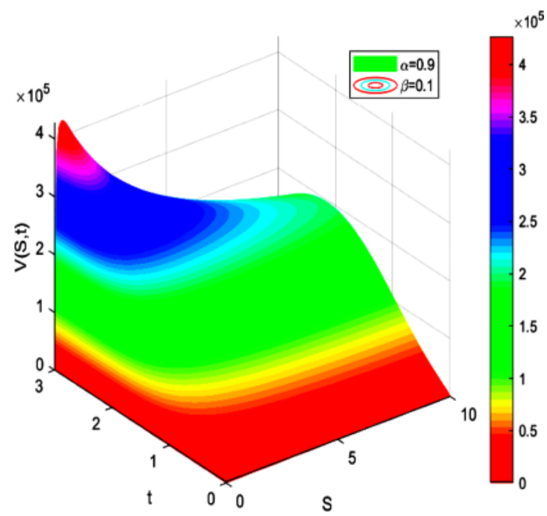


Fig. 51 Plot $V = V(S, t, 0.9, 0.1)$ for $\zeta = 2$

call option increases considerably. This is dangerous owing to the potential of uncapped losses if the stock ascends.

These features illustrate the flexibility and efficiency of BSE in fractal dimensions. The next step is to use numerical simulations and apply our approach to financial markets, such as the double-knockout barrier option, Asian, and seasoned options, among others. Another interesting idea is to examine the solutions and performance of the BSE in fractal dimensions with regard to the complexity of the variables that we include in it. Moreover, extending this study for alternatives fractal and fractional derivatives operators is motivating since it will allow to compare the outcomes with those obtained using various fractal approaches (Atangana 2017; Batogna and Atangana 2019; He and Ain 2020; He and El-Dib 2021; He and Liu 2023). This work has introduced the basic mathematical and qualitative setups; yet, validating our approach using the real data from financial markets is important. Estimate and

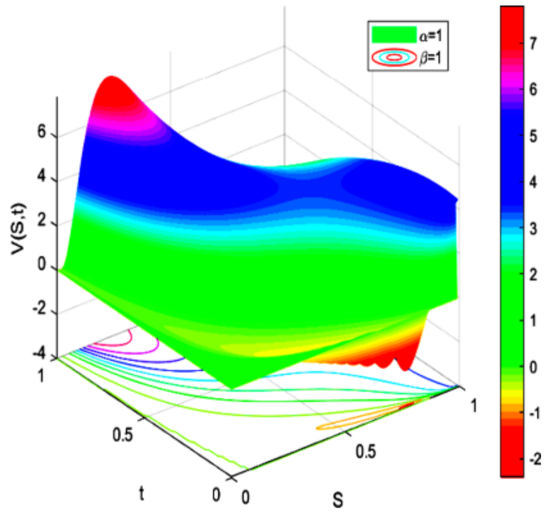


Fig. 52 Plot $V = V(S, t, 1, 1)$ for $\zeta = 2$

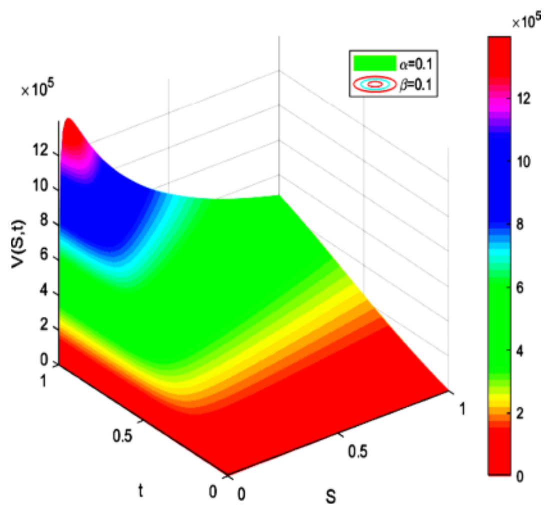


Fig. 53 Plot $V = V(S, t, 0.1, 0.1)$ for $\zeta = 2$

comparing the empirical performance of our fractal option-pricing model with the classical Black–Scholes model, fractional Black Scholes model, and various stochastic financial models found in the literature is also critical. The verification of the significance of the numerical estimates of fractal parameters is also essential. The emergent Black–Scholes models in this study represent, in our opinion, a motivating step to study several financial market problems. We believe that these results may positively contribute to the recent studies planning to overcome the drawbacks in the classical Black–Scholes model. This generalized model represents a new attempt to treat BSE in fractal dimensions and is worth considering for future investigation. Unlike in several research studies, our model is not associated with fractional derivatives, and it offers new analytical results not found within the realm of fractional BSE. In the future, it will be of interest to apply our model to various financial markets models, compare

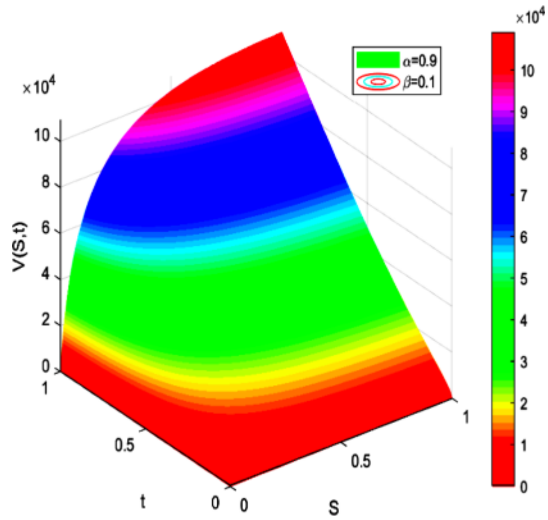


Fig. 54 Plot $V = V(S, t, 0.9, 0.1)$ for $\zeta = 2$

the outcomes with those found in literature, and explore fractal processes, fractional derivatives, and integration as well as their applications to finance. Notably, the fractal differential equations obtained in this study differ from most fractional financial models, which makes it difficult to compare them with others. Further study would be required to determine whether our model is basically comparable to other fractional financial models, although, based on our results, we believe that it generates new dynamics and features. It shows a hint that these financial systems exhibit interesting properties worth studying further. The mathematical languages of these two fields (fractal calculus and fractional calculus) are rather different, but both exhibit motivating outcomes that deserve to be compared in a future work by treating option-pricing models. Additional points that deserve to be explored are financial models with long memory volatility owing to the relevance of memory effects in stock-price dynamics (Fedotov and Tan 2005; Garzarelli et al 2014; Niguyen et al 2020; Tarasov 2020, 2024; Tarasov and Tarasova 2021).These studies have used memory kernel to describe a financial system with memory without altering the correlation between market liquidity and volatility. It will be of interest to study these financial options pricing models with memory based on our fractal approach, make use of the nonlocal fractal calculus concept (Banchuin 2023), and compare the outcomes with those obtained from fractional order financial models. Estimating fractal dimensions using real data and real-life financial applications will also present a challenge to overcome.

List of symbols

S	Stock price at the beginning of the time period of the option
K	Strike price
T	Expiration time
t	Start date
r	Risk-free interest rate
σ	Volatility of the stock
D	Dividend payout
$V(T,S)_{t=T} = (S(T) - K)^{+} = \max(S(T) - K, 0)$	The dividend payout
$Z = V(t, S(t))$ where $V : \mathbb{R}_{+} \times \mathbb{R} \to \mathbb{R}$ a C^2-function	Stochastic process.

$\overline{\nabla}^{\alpha} f(x) := (x/x_0)^{1-\alpha}(\nabla + ((1-\alpha)/2x))f(x)$ with x_0 is a spatial parameter	Fractal gradient of order $\alpha \in (0,1)$
$T_\beta(f)(t) = lim_{\varepsilon \to 0}(f(t+\varepsilon t^{1-\beta}) - f(t))/\varepsilon$	Conformable derivative of order $\beta \in (0,1)$
$\mathbf{N}(\lambda) = \int_{-\infty}^{\lambda} e^{-s^2/2} ds \Big/ \sqrt{2\pi} = (erf(\lambda \Big/ \sqrt{2}) + 1) \Big/ 2$	Cumulative distribution function for the standard normal distribution

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Competing interests
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